

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/235054307>

AFGL Atmospheric Constituent Profiles (0.120km)

Article · May 1986

CITATIONS

893

READS

5,313

5 authors, including:



Shepard Anthony Clough

Clough Radiation Associates, Lexington, MA, United States

211 PUBLICATIONS 28,173 CITATIONS

[SEE PROFILE](#)



Eric Shettle

178 PUBLICATIONS 9,992 CITATIONS

[SEE PROFILE](#)

12

AFGL-TR-86-0110
ENVIRONMENTAL RESEARCH PAPERS, NO. 954

AD-A175 173

AFGL Atmospheric Constituent Profiles (0-120km)

G.P. ANDERSON
S.A. CLOUGH
F.X. KNEIZYS

J.H. CHETWYND
E.P. SHETTLE



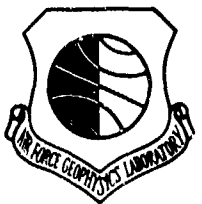
15 May 1986



Approved for public release; distribution unlimited.



DTIC
ELECTE
DEC 18 1986
S B



OPTICAL PHYSICS DIVISION

PROJECT 7670

AIR FORCE GEOPHYSICS LABORATORY

HANSCOM AFB, MA 01731

DTIC FILE COPY

UNCLASSIFIED

SECURITY CLASSIFICATION OF THIS PAGE

REPORT DOCUMENTATION PAGE

1a. REPORT SECURITY CLASSIFICATION UNCLASSIFIED			1b. RESTRICTIVE MARKINGS	
2a. SECURITY CLASSIFICATION AUTHORITY			3. DISTRIBUTION / AVAILABILITY OF REPORT Approved for Public Release; Distribution Unlimited	
2b. DECLASSIFICATION / DOWNGRADING SCHEDULE				
4. PERFORMING ORGANIZATION REPORT NUMBER(S) ERP, No. 954 AFGL-TR-86-0110			5. MONITORING ORGANIZATION REPORT NUMBER(S)	
6a. NAME OF PERFORMING ORGANIZATION Air Force Geophysics Laboratory		6b. OFFICE SYMBOL (if applicable) OPI	7a. NAME OF MONITORING ORGANIZATION	
6c. ADDRESS (City, State, and ZIP Code) Hanscom AFB Massachusetts 01731			7b. ADDRESS (City, State, and ZIP Code)	
8a. NAME OF FUNDING / SPONSORING ORGANIZATION		8b. OFFICE SYMBOL (if applicable)	9. PROCUREMENT INSTRUMENT IDENTIFICATION NUMBER	
8c. ADDRESS (City, State, and ZIP Code)			10. SOURCE OF FUNDING NUMBERS	
			PROGRAM ELEMENT NO. 62101F	PROJECT NO. 7670
			TASK NO. 09	WORK UNIT ACCESSION NO. 76700908
11. TITLE (Include Security Classification) AFGL Atmospheric Constituent Profiles (0-120km)				
12. PERSONAL AUTHOR(S) G.P. ANDERSON, S.A. CLOUGH, F.X. KNEIZYS, J.H. CHETWYND, E.P. SHETTL				
13a. TYPE OF REPORT SCIENTIFIC INTERIM		13b. TIME COVERED FROM TO	14. DATE OF REPORT (Year, Month, Day) 1986 May 15	15. PAGE COUNT 48
16. SUPPLEMENTARY NOTATION				
17. COSATI CODES			18. SUBJECT TERMS (Continue on reverse if necessary and identify by block number)	
FIELD	GROUP	SUB-GROUP	→ ATMOSPHERIC CONSTITUENTS; TEMPERATURE PROFILES; MODEL ATMOSPHERES. ←	
19. ABSTRACT (Continue on reverse if necessary and identify by block number)				
→ An atmospheric data base consisting of volume mixing ratios (0-120km) for twenty-eight (28) minor and trace gases has been assembled for use with spectral radiance/transmittance models. Six reference atmospheres, each defining temperature, pressure and density as a function of altitude (selected from the U.S. Standard Supplements, 1966 and the U.S. Standard Atmosphere, 1976) provide a range of climatological choices. Analogous zonal-mean descriptions for H₂O, O₃, N₂O, CO, and CH₄ have been subsequently adapted from satellite data and/or dynamical-photochemical analyses. The remaining species are defined by single profiles, usually appropriate for U.S. Standard conditions. Because the entire profile set is preferentially based on available measurements, explicit photochemical consistency between the different species has not been maintained. <i>Keywords:</i>				
20. DISTRIBUTION / AVAILABILITY OF ABSTRACT ☐ UNCLASSIFIED/UNLIMITED ☐ SAME AS RPT. ☒ DTIC USERS			21. ABSTRACT SECURITY CLASSIFICATION UNCLASSIFIED	
22a. NAME OF RESPONSIBLE INDIVIDUAL GAIL P. ANDERSON			22b. TELEPHONE (Include Area Code) (617) 377-2335	22c. OFFICE SYMBOL OPI

DD FORM 1473, 84 MAR

83 APR edition may be used until exhausted.
All other editions are obsolete.

SECURITY CLASSIFICATION OF THIS PAGE

UNCLASSIFIED

Contents

1. INTRODUCTION	1
2. ATMOSPHERIC PROFILE DESCRIPTION	3
3. ERROR/VARIABILITY ESTIMATES	16
4. LIMITATIONS	16
REFERENCES	19
APPENDIX A: ATMOSPHERIC PROFILES	21-35
Part I: Reference Atmospheric Model Profiles	
Temperature	23
Density	24
Pressure	25
H ₂ O	26
O ₃	27
N ₂ O	28
CO	29
CH ₄	30
Part II: Constituent Profiles	
H ₂ O, CO ₂ , O ₃ , N ₂ O	32
CO, CH ₄ , O ₂ , NO	32
SO ₂ , NO ₂ , NH ₃ , HNO ₃	33
OH, HF, HCl, HBr	33
HI, ClO, OCS, H ₂ CO	34
HOCl, N ₂ , HCN, CH ₃ Cl	34
H ₂ O ₂ , C ₂ H ₂ , C ₂ H ₆ , PH ₃	35
APPENDIX B: BIBLIOGRAPHY	
Part I. Subject Listing	37
Part II. Alphabetical Listing	39

1. AFGL U.S. STD. Profiles (H_2O , CO_2 , O_3 , N_2O , CO , CH_4 , O_2)

10

Tables

1. Reference Atmospheric Model Profiles, including $T(K)$, $p(mb)$, $n(cm^3)$, and mixing ratios (ppmv) for H_2O , O_3 , N_2O , CO , and CH_4

a. Model 1. Tropical	4
b. Model 2. Midlatitude Summer	5
c. Model 3. Midlatitude Winter	6
d. Model 4. Subarctic Summer	7
e. Model 5. Subarctic Winter	8
f. Model 6. U.S. Standard	9

2. Constituent Profiles (ppmv) for:

a. H_2O , CO_2 , N_2O , CO , CH_4 , O_2	11
b. NO , SO_2 , NO_2 , NH_3 , HNO_3 , OH , HF	12
c. HCl , HBr , HI , ClO , OCS , H_2CO , $HOCl$	13
d. N_2 , HCN , CH_3Cl , H_2O_2 , C_2H_2 , C_2H_6 , PH_3	14



Accession For	
NTIS GRA&I	☒
DTIC TAB	☐
Unannounced	☐
Justification	
By	
Distribution/	
Availability Codes	
Dist	Avail and/or Special
A-1	

DTIC
ELECTE
DEC 18 1986
S B D

AFGL Atmospheric Constituent Profiles (0-120km)

1. INTRODUCTION

Atmospheric radiance-transmittance spectral modeling requires an adequate description of the local thermal and constituent environment. A data base consisting of realistic vertical profiles for temperature and gas mixing ratios has been designed expressly for incorporation into such models. Its thermal structure is represented by a subset of the 1966 Atmospheric Supplements¹ (tropical (15N), middle latitude (45N) summer and winter, subarctic (60N) summer and winter) and the U.S. Standard Model Atmosphere, 1976². The accompanying volume mixing ratio profiles rely as much as possible on current measurements and/or theoretical predictions (see Appendix B).

(Received for publication 14 May 1986)

1. NASA (1966), U.S. Standard Atmosphere Supplements, 1966, U.S. Government Printing Office, Washington, DC.
2. NASA (1976), U.S. Standard Atmosphere Supplements, 1976, U.S. Government Printing Office, Washington, DC.

More extensive literature reviews of atmospheric structure, variability, dynamics and chemistry are available (for example, Smith,³ WMO,^{4a, 4b} and Brasseur and Solomon⁵).

This compilation includes only those gases currently part of the AFGL molecular line parameter atlases^{6, 7}. The range of tabulated atmospheric values for water vapor (H_2O), ozone (O_3), nitrous oxide (N_2O), and methane (CH_4) are primarily inferred from global satellite measurements^{8, 9, 10}. The carbon monoxide (CO) seasonal profiles, however, rely on the predictions of a photochemical-dynamical model¹¹. The remaining individual gas profiles have been derived from a variety of sources. All have been edited to produce the final tabulations; in most cases this consists of smoothing and interpolation to standard altitude levels. Some species, however, require additional extrapolation because of the unavailability of suitable data (particularly above the stratopause). In general, dayside estimates for diurnally varying species (O_3 , NO , and NO_2 , for example) have been adopted.

-
3. Smith, M. A. H. (1982) Compilation of Atmospheric Gas Concentration Profiles From 0-50 km, NASA Tech Mem 83289.
 - 4a. W. M. O. (1982) The Stratosphere 1981: Theory and Measurements, Report No. 11, NASA, Greenbelt, MD.
 - 4b. W. M. O. (1986) Atmospheric Ozone 1985: Assessment of Our Understanding of the Processes Controlling its Present Distribution and Change, WMO Report No. 16, W. M. O., Geneva, Switzerland.
 5. Brasseur, G. and Solomon, S. (1984) Aeronomy of the Middle Atmosphere, D. Reidel Publishing Co, Dordrecht, Holland, Chapter 5: Composition & Chemistry, 440 pp.
 6. Rothman, L.S., Goldman, A., Gillis, J.R., Gamache, R.R., Pickett, H.M., Poynter, R.L., Husson, N. and Chedin, A. (1983b), AFGL trace gas compilation: 1982 version, Appl. Opt., 22, 1616-1627.
 7. Rothman, L.S., Gamache, R.R., Barbe, A., Goldman, A., Gillis, H.R., Brown, L.A., Toth, R.A., Flaud, J.-M. and Camy-Peyret, C. (1983a), AFGL atmospheric absorption line parameters compilation: 1982 edition, Appl. Opt., 22, 2247-2256.
 8. Russell III, J.M., Gille, J.C., Remsberg, E.E., Gordley, L.L., Baily, P.L., Fischer, H., Girard, A., Drayon, S.R., Evans, W.F.J. and Harries, J.E. (1984b) Validation of water vapor results measured by the LIMS experiment on Nimbus 7, J. Geophys. Res., 89, 5115-5120.
 9. Keating, G.M. and Young D.F. (1985) Interim reference ozone models for the middle atmosphere, Interim COSPAR Reference Atmosphere for Altitudes 20-120km, in preparation.
 10. Jones, R.L. and Pyle, J.A. (1984) Observations of CH_4 and N_2O by the Nimbus 7 SAMS: A comparison with in situ data and two-dimensional numerical model calculations, J. Geophys. Res., 89, 5263-5379.
 11. Solomon, S., Garcia, R.R., Olivero, J.J., Bevilacqua, R.M., Schwartz, P.R., Clancy, R.T. and Mahleman, D.O. (1985) Photochemistry and transport of CO in the middle atmosphere, J. Atmos. Sci., 42, 1072-1083.

2. ATMOSPHERIC PROFILE DESCRIPTION

The six reference atmospheres, each with associated volume mixing ratio profiles for H_2O , O_3 , N_2O , CO , and CH_4 , are presented in Table 1. Along with CO_2 , these are the most radiatively active molecules. Sample profiles, appropriate for the U. S. Standard atmosphere conditions (Model 6), are shown in Figure 1. Because the mixing ratios of CO_2 and O_2 have been held seasonally invariant, they are listed in Table 2, as part of a set of single profiles numbered according to the AFGL Line Atlas^{6, 7} numbering system; i.e. H_2O is 1, CO_2 is 2, etc. The first seven molecules in Table 2 are the same as those in Figure 1 for the U.S. Standard Atmosphere. The 21 additional species, as identified on the AFGL trace gas compilation⁷, are; NO , SO_2 , NO_2 , NH_3 , HNO_3 , OH , HF , HCl , HBr , HI , ClO , OCS , H_2CO , HOCl , N_2 , HCN , CH_3Cl , H_2O_2 , C_2H_2 , C_2H_6 , and PH_3 . Graphical representations for all the tabular data are available in Appendix A. (Note: although N_2 is the dominant atmospheric gas, it appears as only a trace spectral contributor.)

Table 1. Reference Model Atmospheric Profiles

Model = 1	Tropic (15N Annual Average)
Model = 2	Mid-Latitude Summer (45N July)
Model = 3	Mid-Latitude Winter (45N Jan)
Model = 4	Sub Arctic Summer (60N July)
Model = 5	Sub-Arctic Winter (60N Jan)
Model = 6	U.S. Standard (1976)

This tabular presentation includes: Altitude (km), Pressure (mb), Density (cm^{-3}), and mixing ratios (ppmv) for H_2O , O_3 , N_2O , CO , and CH_4 . Profiles for CO_2 and O_2 can be found in Table 2.

[(*) indicates subsequent extrapolation adopted for that species]

Table 1a. Reference Atmospheric Model Profiles, Model 1. Tropical

MODEL = 1 TROPICAL								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM^{-3})	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.013E+03	299.7	2.450E+19	2.59E+04	2.87E-02	3.20E-01	1.50E-01	1.70E+00
1.00	9.040E+02	293.7	2.231E+19	1.95E+04	3.15E-02	3.20E-01	1.45E-01	1.70E+00
2.00	8.050E+02	287.7	2.028E+19	1.53E+04	3.34E-02	3.20E-01	1.40E-01	1.70E+00
3.00	7.150E+02	283.7	1.827E+19	8.60E+03	3.50E-02	3.20E-01	1.35E-01	1.70E+00
4.00	6.330E+02	277.0	1.656E+19	4.44E+03	3.56E-02	3.20E-01	1.31E-01	1.70E+00
5.00	5.590E+02	270.3	1.499E+19	3.35E+03	3.77E-02	3.20E-01	1.30E-01	1.70E+00
6.00	4.920E+02	263.6	1.353E+19	2.10E+03	3.99E-02	3.20E-01	1.29E-01	1.70E+00
7.00	4.320E+02	257.0	1.218E+19	1.29E+03	4.22E-02	3.20E-01	1.25E-01	1.70E+00
8.00	3.780E+02	250.3	1.095E+19	7.64E+02	4.47E-02	3.20E-01	1.19E-01	1.70E+00
9.00	3.290E+02	243.6	9.789E+18	4.10E+02	5.00E-02	3.20E-01	1.09E-01	1.69E+00
10.00	2.860E+02	237.0	8.747E+18	1.91E+02	5.60E-02	3.18E-01	9.96E-02	1.69E+00
11.00	2.470E+02	230.1	7.780E+18	7.31E+01	6.61E-02	3.14E-01	8.96E-02	1.68E+00
12.00	2.130E+02	223.6	6.904E+18	2.91E+01	7.82E-02	3.10E-01	7.81E-02	1.66E+00
13.00	1.820E+02	217.0	6.079E+18	9.90E+00	9.29E-02	3.05E-01	6.37E-02	1.65E+00
14.00	1.560E+02	210.3	5.377E+18	6.22E+00	1.05E-01	3.00E-01	5.03E-02	1.63E+00
15.00	1.320E+02	203.7	4.697E+18	4.00E+00	1.26E-01	2.94E-01	3.94E-02	1.61E+00
16.00	1.110E+02	197.0	4.084E+18	3.00E+00	1.44E-01	2.88E-01	3.07E-02	1.58E+00
17.00	9.370E+01	194.8	3.486E+18	2.90E+00	2.50E-01	2.78E-01	2.49E-02	1.55E+00
18.00	7.890E+01	198.8	2.877E+18	2.75E+00	5.00E-01	2.67E-01	1.97E-02	1.52E+00
19.00	6.660E+01	202.7	2.381E+18	2.60E+00	9.50E-01	2.53E-01	1.55E-02	1.48E+00
20.00	5.650E+01	206.7	1.981E+18	2.60E+00	1.40E+00	2.37E-01	1.33E-02	1.42E+00
21.00	4.800E+01	210.7	1.651E+18	2.65E+00	1.80E+00	2.19E-01	1.23E-02	1.36E+00
22.00	4.090E+01	214.6	1.381E+18	2.80E+00	2.40E+00	2.05E-01	1.23E-02	1.27E+00
23.00	3.500E+01	217.0	1.169E+18	2.90E+00	3.40E+00	1.97E-01	1.31E-02	1.19E+00
24.00	3.000E+01	219.2	9.920E+17	3.20E+00	4.30E+00	1.88E-01	1.40E-02	1.12E+00
25.00	2.570E+01	221.4	8.413E+17	3.25E+00	5.40E+00	1.76E-01	1.52E-02	1.06E+00
27.50	1.763E+01	227.0	5.629E+17	3.60E+00	7.80E+00	1.59E-01	1.72E-02	9.87E-01
30.00	1.220E+01	232.3	3.807E+17	4.00E+00	9.30E+00	1.42E-01	2.00E-02	9.14E-01
32.50	8.520E+00	237.7	2.598E+17	4.30E+00	9.85E+00	1.17E-01	2.27E-02	8.30E-01
35.00	6.000E+00	243.1	1.789E+17	4.60E+00	9.70E+00	9.28E-02	2.49E-02	7.46E-01
37.50	4.260E+00	248.5	1.243E+17	4.90E+00	8.80E+00	6.69E-02	2.74E-02	6.62E-01
40.00	3.050E+00	254.0	8.703E+16	5.20E+00	7.50E+00	4.51E-02	3.10E-02	5.64E-01
42.50	2.200E+00	259.4	6.147E+16	5.50E+00	5.90E+00	2.75E-02	3.51E-02	4.61E-01
45.00	1.590E+00	264.8	4.352E+16	5.70E+00	4.50E+00	1.59E-02	3.99E-02	3.63E-01
47.50	1.160E+00	269.6	3.119E+16	5.90E+00	3.45E+00	9.38E-03	4.48E-02	2.77E-01
50.00	8.540E-01	270.2	2.291E+16	6.00E+00	2.80E+00	4.75E-03*	5.09E-02	2.10E-01
55.00	4.560E-01	263.4	1.255E+16	6.00E+00	1.80E+00	3.00E-03	5.99E-02	1.65E-01
60.00	2.390E-01	253.1	6.844E+15	6.00E+00	1.10E+00	2.07E-03	6.96E-02	1.50E-01
65.00	1.210E-01	236.0	3.716E+15	5.40E+00	6.50E-01	1.51E-03	9.19E-02	1.50E-01
70.00	5.800E-02	218.9	1.920E+15	4.50E+00	3.00E-01	1.15E-03	1.94E-01	1.50E-01
75.00	2.600E-02	201.8	9.338E+14	3.30E+00	1.80E-01	8.89E-04	5.69E-01	1.50E-01
80.00	1.100E-02	184.8	4.314E+14	2.10E+00	3.30E-01	7.06E-04	1.55E+00	1.50E-01
85.00	4.400E-03	177.1	1.801E+14	1.30E+00	5.00E-01	5.72E-04	3.85E+00	1.50E-01
90.00	1.720E-03	177.0	7.043E+13	8.50E-01	5.20E-01	4.71E-04	6.59E+00	1.40E-01
95.00	6.880E-04	184.3	2.706E+13	5.40E-01	5.00E-01	3.93E-04	1.04E+01	1.30E-01
100.00	2.890E-04	190.7	1.098E+13	4.00E-01	4.00E-01	3.32E-04	1.71E+01	1.20E-01
105.00	1.300E-04	212.0	4.445E+12	3.40E-01	2.00E-01	2.84E-04	2.47E+01	1.10E-01
110.00	6.470E-05	241.6	1.941E+12	2.80E-01	5.00E-02	2.44E-04	3.36E+01	9.50E-02
115.00	3.600E-05	299.7	8.706E+11	2.40E-01	5.00E-03	2.12E-04	4.15E+01	6.00E-02
120.00	2.250E-05	380.0	4.225E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

Table 1b. Reference Atmospheric Model Profiles, Model 2. Midlatitude Summer

MODEL = 2 MIDLATITUDE SUMMER								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM-3)	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.013E+03	284.2	2.496E+19	1.88E+04	3.02E-02	3.20E-01	1.50E-01	1.70E+00
1.00	9.020E+02	289.7	2.257E+19	1.38E+04	3.34E-02	3.20E-01	1.45E-01	1.70E+00
2.00	8.020E+02	285.2	2.038E+19	9.68E+03	3.69E-02	3.20E-01	1.40E-01	1.70E+00
3.00	7.100E+02	279.2	1.843E+19	5.98E+03	4.22E-02	3.20E-01	1.35E-01	1.70E+00
4.00	6.280E+02	273.2	1.668E+19	3.81E+03	4.82E-02	3.20E-01	1.31E-01	1.70E+00
5.00	5.540E+02	267.2	1.503E+19	2.23E+03	5.51E-02	3.20E-01	1.30E-01	1.69E+00
6.00	4.870E+02	261.2	1.351E+19	1.51E+03	6.41E-02	3.20E-01	1.29E-01	1.67E+00
7.00	4.260E+02	254.7	1.212E+19	1.02E+03	7.76E-02	3.20E-01	1.25E-01	1.65E+00
8.00	3.720E+02	248.2	1.086E+19	6.46E+02	9.13E-02	3.20E-01	1.19E-01	1.63E+00
9.00	3.240E+02	241.7	9.716E+18	4.13E+02	1.11E-01	3.16E-01	1.09E-01	1.62E+00
10.00	2.810E+02	235.3	8.656E+18	2.47E+02	1.30E-01	3.10E-01	9.96E-02	1.58E+00
11.00	2.430E+02	228.8	7.698E+18	9.56E+01	1.79E-01	2.99E-01	8.96E-02	1.54E+00
12.00	2.090E+02	222.3	6.814E+18	2.94E+01	2.23E-01	2.94E-01	7.81E-02	1.51E+00
13.00	1.790E+02	215.8	6.012E+18	8.00E+00	3.00E-01	2.86E-01	6.37E-02	1.48E+00
14.00	1.530E+02	215.7	5.141E+18	5.00E+00	4.40E-01	2.80E-01	5.03E-02	1.45E+00
15.00	1.300E+02	215.7	4.368E+18	3.40E+00	5.00E-01	2.72E-01	3.94E-02	1.42E+00
16.00	1.110E+02	215.7	3.730E+18	3.30E+00	6.00E-01	2.61E-01	3.07E-02	1.39E+00
17.00	9.500E+01	215.7	3.192E+18	3.20E+00	7.00E-01	2.42E-01	2.49E-02	1.36E+00
18.00	8.120E+01	216.8	2.715E+18	3.15E+00	1.00E+00	2.17E-01	1.97E-02	1.32E+00
19.00	6.950E+01	217.9	2.312E+18	3.20E+00	1.50E+00	1.84E-01	1.55E-02	1.28E+00
20.00	5.950E+01	219.2	1.967E+18	3.30E+00	2.00E+00	1.61E-01	1.33E-02	1.22E+00
21.00	5.100E+01	220.4	1.677E+18	3.45E+00	2.40E+00	1.32E-01	1.23E-02	1.15E+00
22.00	4.370E+01	221.6	1.429E+18	3.60E+00	2.90E+00	1.15E-01	1.23E-02	1.07E+00
23.00	3.760E+01	222.8	1.223E+18	3.85E+00	3.40E+00	1.04E-01	1.31E-02	9.73E-01
24.00	3.220E+01	223.9	1.042E+18	4.00E+00	4.00E+00	9.62E-02	1.40E-02	8.80E-01
25.00	2.770E+01	225.1	8.919E+17	4.20E+00	4.80E+00	8.96E-02	1.52E-02	7.89E-01
27.50	1.907E+01	228.5	6.050E+17	4.45E+00	6.00E+00	8.01E-02	1.72E-02	7.05E-01
30.00	1.320E+01	233.7	4.094E+17	4.70E+00	7.00E+00	6.70E-02	2.00E-02	6.32E-01
32.50	9.300E+00	239.0	2.820E+17	4.85E+00	8.10E+00	4.96E-02	2.27E-02	5.59E-01
35.00	6.520E+00	245.2	1.927E+17	4.95E+00	8.90E+00	3.70E-02	2.49E-02	5.01E-01
37.50	4.640E+00	251.3	1.338E+17	5.00E+00	8.70E+00	2.52E-02	2.72E-02	4.45E-01
40.00	3.330E+00	257.5	9.373E+16	5.10E+00	7.55E+00	1.74E-02	2.96E-02	3.92E-01
42.50	2.410E+00	263.7	6.624E+16	5.30E+00	5.90E+00	1.16E-02	3.14E-02	3.39E-01
45.00	1.760E+00	269.9	4.726E+16	5.45E+00	4.50E+00	7.67E-03	3.31E-02	2.87E-01
47.50	1.290E+00	275.2	3.398E+16	5.50E+00	3.50E+00	5.32E-03	3.49E-02	2.38E-01
50.00	9.510E-01	275.7	2.500E+16	5.50E+00	2.80E+00	3.22E-03	3.65E-02	1.94E-01
55.00	5.150E-01	269.3	1.386E+16	5.35E+00	1.80E+00	2.03E-03	3.92E-02	1.57E-01
60.00	2.720E-01	257.1	7.668E+15	5.00E+00	1.30E+00	1.40E-03	4.67E-02	1.50E-01
65.00	1.390E-01	240.1	4.196E+15	4.40E+00	8.00E-01	1.02E-03	6.40E-02	1.50E-01
70.00	6.700E-02	218.1	2.227E+15	3.70E+00	4.00E-01	7.77E-04	1.18E-01	1.50E-01
75.00	3.000E-02	198.1	1.109E+15	2.95E+00	1.90E-01	6.26E-04	2.94E-01	1.50E-01
80.00	1.200E-02	174.1	4.996E+14	2.10E+00	2.00E-01	5.17E-04	6.82E-01	1.50E-01
85.00	4.480E-03	165.1	1.967E+14	1.33E+00	5.70E-01	4.35E-04	1.47E+00	1.50E-01
90.00	1.640E-03	165.0	7.204E+13	8.50E-01	7.50E-01	3.73E-04	2.85E+00	1.40E-01
95.00	6.250E-04	178.3	2.541E+13	5.40E-01	7.00E-01	3.24E-04	5.17E+00	1.30E-01
100.00	2.580E-04	190.5	9.816E+12	4.00E-01	4.00E-01	2.84E-04	1.01E+01	1.20E-01
105.00	1.170E-04	222.2	3.816E+12	3.40E-01	2.00E-01	2.52E-04	1.87E+01	1.10E-01
110.00	6.110E-05	262.4	1.688E+12	2.80E-01	5.00E-02	2.26E-04	2.86E+01	9.50E-02
115.00	3.560E-05	316.8	8.145E+11	2.40E-01	5.00E-03	2.04E-04	3.89E+01	6.00E-02
120.00	2.270E-05	380.0	4.330E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

Table 1c. Reference Atmospheric Model Profiles, Model 3. Midlatitude Winter

MODEL = 3 MIDLATITUDE WINTER								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM-3)	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.018E+03	272.2	2.711E+19	4.32E+03	2.78E-02	3.20E-01	1.50E-01	1.70E+00
1.00	8.973E+02	268.7	2.420E+19	3.45E+03	2.80E-02	3.20E-01	1.45E-01	1.70E+00
2.00	7.897E+02	265.2	2.158E+19	2.79E+03	2.85E-02	3.20E-01	1.40E-01	1.70E+00
3.00	6.938E+02	261.7	1.922E+19	2.09E+03	3.20E-02	3.20E-01	1.35E-01	1.70E+00
4.00	6.081E+02	255.7	1.724E+19	1.28E+03	3.57E-02	3.20E-01	1.31E-01	1.70E+00
5.00	5.313E+02	249.7	1.542E+19	8.24E+02	4.72E-02	3.20E-01	1.30E-01	1.69E+00
6.00	4.627E+02	243.7	1.376E+19	5.10E+02	5.84E-02	3.20E-01	1.29E-01	1.67E+00
7.00	4.016E+02	237.7	1.225E+19	2.32E+02	7.89E-02	3.20E-01	1.25E-01	1.65E+00
8.00	3.473E+02	231.7	1.086E+19	1.08E+02	1.04E-01	3.20E-01	1.19E-01	1.63E+00
9.00	2.993E+02	225.7	9.612E+18	5.57E+01	1.57E-01	3.16E-01	1.09E-01	1.62E+00
10.00	2.568E+02	219.7	8.472E+18	2.96E+01	2.37E-01	3.10E-01	9.96E-02	1.58E+00
11.00	2.199E+02	219.2	7.271E+18	1.00E+01	3.62E-01	2.99E-01	8.96E-02	1.54E+00
12.00	1.882E+02	218.7	6.237E+18	6.00E+00	5.23E-01	2.94E-01	7.81E-02	1.51E+00
13.00	1.611E+02	218.2	5.351E+18	5.00E+00	7.04E-01	2.86E-01	6.37E-02	1.48E+00
14.00	1.378E+02	217.7	4.588E+18	4.80E+00	8.00E-01	2.80E-01	5.03E-02	1.45E+00
15.00	1.178E+02	217.2	3.931E+18	4.70E+00	9.00E-01	2.72E-01	3.94E-02	1.42E+00
16.00	1.007E+02	216.7	3.366E+18	4.60E+00	1.10E+00	2.61E-01	3.07E-02	1.39E+00
17.00	8.610E+01	216.2	2.886E+18	4.50E+00	1.40E+00	2.42E-01	2.49E-02	1.36E+00
18.00	7.360E+01	215.7	2.473E+18	4.50E+00	1.80E+00	2.17E-01	1.97E-02	1.32E+00
19.00	6.280E+01	215.2	2.115E+18	4.50E+00	2.30E+00	1.84E-01	1.55E-02	1.28E+00
20.00	5.370E+01	215.2	1.809E+18	4.50E+00	2.90E+00	1.62E-01	1.33E-02	1.22E+00
21.00	4.580E+01	215.2	1.543E+18	4.50E+00	3.50E+00	1.36E-01	1.23E-02	1.15E+00
22.00	3.910E+01	215.2	1.317E+18	4.53E+00	3.90E+00	1.23E-01	1.23E-02	1.07E+00
23.00	3.340E+01	215.2	1.125E+18	4.55E+00	4.30E+00	1.12E-01	1.31E-02	9.73E-01
24.00	2.860E+01	215.2	9.633E+17	4.60E+00	4.70E+00	1.05E-01	1.40E-02	8.80E-01
25.00	2.440E+01	215.2	8.218E+17	4.65E+00	5.10E+00	9.66E-02	1.50E-02	7.93E-01
27.50	1.646E+01	215.5	5.536E+17	4.70E+00	5.60E+00	8.69E-02	1.60E-02	7.13E-01
30.00	1.110E+01	217.4	3.701E+17	4.75E+00	6.10E+00	7.52E-02	1.71E-02	6.44E-01
32.50	7.560E+00	220.4	2.486E+17	4.80E+00	6.80E+00	6.13E-02	1.85E-02	5.75E-01
35.00	5.180E+00	227.9	1.647E+17	4.85E+00	7.10E+00	5.12E-02	2.00E-02	5.05E-01
37.50	3.600E+00	235.5	1.108E+17	4.90E+00	7.20E+00	3.97E-02	2.15E-02	4.48E-01
40.00	2.530E+00	243.2	7.540E+16	4.95E+00	6.90E+00	3.00E-02	2.33E-02	3.93E-01
42.50	1.800E+00	250.8	5.202E+16	5.00E+00	5.90E+00	2.08E-02	2.62E-02	3.40E-01
45.00	1.290E+00	258.5	3.617E+16	5.00E+00	4.60E+00	1.31E-02	3.08E-02	2.88E-01
47.50	9.400E-01	265.1	2.570E+16	5.00E+00	3.70E+00	8.07E-03	3.80E-02	2.39E-01
50.00	6.830E-01	265.7	1.863E+16	4.95E+00	2.75E+00	4.16E-03	6.25E-02	1.94E-01
55.00	3.620E-01	260.6	1.007E+16	4.85E+00	1.70E+00	2.63E-03	1.48E-01	1.57E-01
60.00	1.880E-01	250.8	5.433E+15	4.50E+00	1.00E+00	1.81E-03	2.93E-01	1.50E-01
65.00	9.500E-02	240.9	2.858E+15	4.00E+00	5.50E-01	1.32E-03	5.59E-01	1.50E-01
70.00	4.700E-02	230.7	1.477E+15	3.30E+00	3.20E-01	1.01E-03	1.08E+00	1.50E-01
75.00	2.220E-02	220.4	7.301E+14	2.70E+00	2.50E-01	7.88E-04	1.90E+00	1.50E-01
80.00	1.030E-02	210.1	3.553E+14	2.00E+00	2.30E-01	6.33E-04	2.96E+00	1.50E-01
85.00	4.560E-03	199.8	1.654E+14	1.33E+00	5.50E-01	5.19E-04	4.53E+00	1.50E-01
90.00	1.980E-03	199.5	7.194E+13	8.50E-01	8.00E-01	4.33E-04	6.86E+00	1.40E-01
95.00	8.770E-04	208.3	3.052E+13	5.40E-01	8.00E-01	3.67E-04	1.05E+01	1.30E-01
100.00	4.074E-04	218.6	1.351E+13	4.00E-01	4.00E-01	3.14E-04	1.71E+01	1.20E-01
105.00	2.000E-04	237.1	6.114E+12	3.40E-01	2.00E-01	2.72E-04	2.47E+01	1.10E-01
110.00	1.057E-04	259.5	2.952E+12	2.80E-01	5.00E-02	2.37E-04	3.36E+01	9.50E-02
115.00	5.980E-05	293.0	1.479E+12	2.40E-01	5.00E-03	2.09E-04	4.15E+01	6.00E-02
120.00	3.600E-05	333.0	7.836E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

Table 1d. Reference Atmospheric Model Profiles, Model 4. Subarctic Summer

MODEL = 4 SUBARCTIC SUMMER								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM-3)	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.010E+03	287.2	2.549E+19	1.19E+04	2.41E-02	3.10E-01	1.50E-01	1.70E+00
1.00	8.960E+02	281.7	2.305E+19	8.70E+03	2.94E-02	3.10E-01	1.45E-01	1.70E+00
2.00	7.829E+02	276.3	2.080E+19	6.75E+03	3.38E-02	3.10E-01	1.40E-01	1.70E+00
3.00	7.000E+02	270.9	1.873E+19	4.82E+03	3.89E-02	3.10E-01	1.35E-01	1.70E+00
4.00	6.160E+02	265.5	1.682E+19	3.38E+03	4.48E-02	3.08E-01	1.31E-01	1.70E+00
5.00	5.410E+02	260.1	1.508E+19	2.22E+03	5.33E-02	3.02E-01	1.30E-01	1.69E+00
6.00	4.740E+02	253.1	1.357E+19	1.33E+03	6.56E-02	2.91E-01	1.29E-01	1.67E+00
7.00	4.130E+02	246.1	1.216E+19	7.97E+02	7.74E-02	2.82E-01	1.25E-01	1.65E+00
8.00	3.590E+02	239.2	1.088E+19	4.00E+02	9.11E-02	2.76E-01	1.19E-01	1.63E+00
9.00	3.108E+02	232.2	9.701E+18	1.30E+02	1.42E-01	2.70E-01	1.09E-01	1.62E+00
10.00	2.677E+02	225.2	8.616E+18	4.24E+01	1.89E-01	2.65E-01	9.96E-02	1.58E+00
11.00	2.300E+02	225.2	7.402E+18	1.33E+01	3.05E-01	2.60E-01	8.96E-02	1.54E+00
12.00	1.977E+02	225.2	6.363E+18	6.00E+00	4.10E-01	2.55E-01	7.81E-02	1.51E+00
13.00	1.700E+02	225.2	5.471E+18	4.45E+00	5.00E-01	2.49E-01	6.37E-02	1.47E+00
14.00	1.460E+02	225.2	4.699E+18	4.00E+00	6.00E-01	2.43E-01	5.03E-02	1.43E+00
15.00	1.260E+02	225.2	4.055E+18	4.00E+00	7.00E-01	2.36E-01	3.94E-02	1.39E+00
16.00	1.080E+02	225.2	3.476E+18	4.00E+00	8.50E-01	2.28E-01	3.07E-02	1.34E+00
17.00	9.280E+01	225.2	2.987E+18	4.05E+00	1.00E+00	2.18E-01	2.49E-02	1.29E+00
18.00	7.980E+01	225.2	2.568E+18	4.30E+00	1.30E+00	2.04E-01	1.97E-02	1.23E+00
19.00	6.860E+01	225.2	2.208E+18	4.50E+00	1.70E+00	1.82E-01	1.55E-02	1.16E+00
20.00	5.900E+01	225.2	1.899E+18	4.80E+00	2.10E+00	1.57E-01	1.33E-02	1.07E+00
21.00	5.070E+01	225.2	1.632E+18	4.70E+00	2.70E+00	1.35E-01	1.23E-02	9.90E-01
22.00	4.360E+01	225.2	1.403E+18	4.80E+00	3.30E+00	1.22E-01	1.23E-02	9.17E-01
23.00	3.750E+01	225.2	1.207E+18	4.83E+00	3.70E+00	1.10E-01	1.31E-02	8.57E-01
24.00	3.228E+01	226.6	1.033E+18	4.85E+00	4.20E+00	9.89E-02	1.40E-02	8.01E-01
25.00	2.780E+01	228.1	8.834E+17	4.90E+00	4.50E+00	8.78E-02	1.51E-02	7.48E-01
27.50	1.923E+01	231.0	6.034E+17	4.95E+00	5.30E+00	7.33E-02	1.65E-02	6.96E-01
30.00	1.340E+01	235.1	4.131E+17	5.00E+00	5.70E+00	5.94E-02	1.81E-02	6.44E-01
32.50	9.400E+00	240.0	2.839E+17	5.00E+00	6.80E+00	4.16E-02	2.00E-02	5.89E-01
35.00	6.610E+00	247.2	1.938E+17	5.00E+00	7.70E+00	3.03E-02	2.18E-02	5.24E-01
37.50	4.720E+00	254.6	1.344E+17	5.00E+00	7.80E+00	1.95E-02	2.34E-02	4.51E-01
40.00	3.400E+00	262.1	9.402E+16	5.00E+00	7.00E+00	1.27E-02	2.50E-02	3.71E-01
42.50	2.480E+00	269.5	6.670E+16	5.00E+00	5.40E+00	9.00E-03	2.65E-02	2.99E-01
45.00	1.820E+00	273.6	4.821E+16	5.00E+00	4.20E+00	6.29E-03	2.81E-02	2.45E-01
47.50	1.340E+00	276.2	3.516E+16	5.00E+00	3.20E+00	4.56E-03	3.00E-02	2.00E-01
50.00	9.870E-01	277.2	2.581E+16	4.95E+00	2.50E+00	2.80E-03	3.22E-02	1.66E-01
55.00	5.370E-01	274.0	1.421E+16	4.85E+00	1.70E+00	1.77E-03	3.65E-02	1.50E-01
60.00	2.880E-01	262.7	7.946E+15	4.50E+00	1.20E+00	1.21E-03	4.59E-02	1.50E-01
65.00	1.470E-01	239.7	4.445E+15	4.00E+00	8.00E-01	8.87E-04	6.38E-02	1.50E-01
70.00	7.100E-02	216.6	2.376E+15	3.30E+00	4.00E-01	6.76E-04	1.18E-01	1.50E-01
75.00	3.200E-02	193.6	1.198E+15	2.70E+00	2.00E-01	5.54E-04	3.03E-01	1.50E-01
80.00	1.250E-02	170.6	5.311E+14	2.00E+00	1.80E-01	4.65E-04	7.89E-01	1.50E-01
85.00	4.510E-03	161.7	2.022E+14	1.33E+00	6.50E-01	3.98E-04	1.82E+00	1.50E-01
90.00	1.610E-03	161.6	7.221E+13	8.50E-01	9.00E-01	3.46E-04	3.40E+00	1.40E-01
95.00	6.060E-04	176.8	2.484E+13	5.40E-01	8.00E-01	3.05E-04	5.92E+00	1.30E-01
100.00	2.480E-04	190.4	9.441E+12	4.00E-01	4.00E-01	2.71E-04	1.04E+01	1.20E-01
105.00	1.130E-04	226.0	3.624E+12	3.40E-01	2.00E-01	2.44E-04	1.88E+01	1.10E-01
110.00	6.000E-05	270.1	1.610E+12	2.80E-01	5.00E-02	2.21E-04	2.87E+01	9.50E-02
115.00	3.540E-05	322.7	7.951E+11	2.40E-01	5.00E-03	2.02E-04	3.89E+01	6.00E-02
120.00	2.260E-05	380.0	4.311E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

Table 1e. Reference Atmospheric Model Profiles, Model 5. Subarctic Winter

MODEL = 5 SUBARCTIC WINTER								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM-3)	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.013E+03	257.2	2.855E+19	1.41E+03	1.80E-02	3.20E-01	1.50E-01	1.70E+00
1.00	8.878E+02	259.1	2.484E+19	1.62E+03	2.07E-02	3.20E-01	1.45E-01	1.70E+00
2.00	7.775E+02	255.9	2.202E+19	1.43E+03	2.34E-02	3.20E-01	1.40E-01	1.70E+00
3.00	6.798E+02	252.7	1.950E+19	1.17E+03	2.77E-02	3.20E-01	1.35E-01	1.70E+00
4.00	5.932E+02	247.7	1.736E+19	7.90E+02	3.25E-02	3.20E-01	1.31E-01	1.70E+00
5.00	5.158E+02	240.9	1.552E+19	4.31E+02	3.80E-02	3.20E-01	1.30E-01	1.69E+00
6.00	4.467E+02	234.1	1.383E+19	2.37E+02	4.45E-02	3.20E-01	1.29E-01	1.67E+00
7.00	3.853E+02	227.3	1.229E+19	1.47E+02	7.25E-02	3.20E-01	1.25E-01	1.65E+00
8.00	3.308E+02	220.6	1.087E+19	3.38E+01	1.04E-01	3.20E-01	1.19E-01	1.63E+00
9.00	2.829E+02	217.2	9.440E+18	2.98E+01	2.10E-01	3.16E-01	1.09E-01	1.62E+00
10.00	2.418E+02	217.2	8.069E+18	2.00E+01	3.00E-01	3.10E-01	9.96E-02	1.58E+00
11.00	2.067E+02	217.2	6.898E+18	1.00E+01	3.50E-01	2.99E-01	8.96E-02	1.54E+00
12.00	1.766E+02	217.2	5.893E+18	6.00E+00	4.00E-01	2.94E-01	7.81E-02	1.51E+00
13.00	1.510E+02	217.2	5.039E+18	4.45E+00	6.50E-01	2.86E-01	6.37E-02	1.47E+00
14.00	1.291E+02	217.2	4.308E+18	4.50E+00	9.00E-01	2.80E-01	5.03E-02	1.43E+00
15.00	1.103E+02	217.2	3.681E+18	4.55E+00	1.20E+00	2.72E-01	3.94E-02	1.39E+00
16.00	9.431E+01	216.6	3.156E+18	4.60E+00	1.50E+00	2.61E-01	3.07E-02	1.34E+00
17.00	8.058E+01	216.0	2.704E+18	4.65E+00	1.90E+00	2.42E-01	2.49E-02	1.29E+00
18.00	6.882E+01	215.4	2.316E+18	4.70E+00	2.45E+00	2.17E-01	1.97E-02	1.23E+00
19.00	5.875E+01	214.8	1.982E+18	4.75E+00	3.10E+00	1.84E-01	1.55E-02	1.16E+00
20.00	5.014E+01	214.2	1.697E+18	4.80E+00	3.70E+00	1.62E-01	1.33E-02	1.08E+00
21.00	4.277E+01	213.6	1.451E+18	4.85E+00	4.00E+00	1.38E-01	1.23E-02	1.01E+00
22.00	3.647E+01	213.0	1.241E+18	4.90E+00	4.20E+00	1.23E-01	1.23E-02	9.56E-01
23.00	3.109E+01	212.4	1.061E+18	4.95E+00	4.50E+00	1.12E-01	1.31E-02	9.01E-01
24.00	2.649E+01	211.8	9.065E+17	5.00E+00	4.60E+00	1.04E-01	1.40E-02	8.48E-01
25.00	2.256E+01	211.2	7.742E+17	5.00E+00	4.70E+00	9.57E-02	1.52E-02	7.96E-01
27.50	1.513E+01	213.6	5.134E+17	5.00E+00	4.90E+00	6.60E-02	1.72E-02	7.45E-01
30.00	1.020E+01	216.0	3.423E+17	5.00E+00	5.40E+00	7.31E-02	2.04E-02	6.94E-01
32.50	6.910E+00	218.5	2.292E+17	5.00E+00	5.90E+00	5.71E-02	2.49E-02	6.43E-01
35.00	4.701E+00	222.3	1.533E+17	5.00E+00	6.20E+00	4.67E-02	3.17E-02	5.88E-01
37.50	3.230E+00	228.5	1.025E+17	5.00E+00	6.25E+00	3.44E-02	4.43E-02	5.24E-01
40.00	2.243E+00	234.7	6.927E+16	5.00E+00	5.90E+00	2.47E-02	6.47E-02	4.51E-01
42.50	1.570E+00	240.8	4.726E+16	5.00E+00	5.10E+00	1.63E-02	1.04E-01	3.71E-01
45.00	1.113E+00	247.0	3.266E+16	5.00E+00	4.10E+00	1.07E-02	1.51E-01	3.00E-01
47.50	7.900E-01	253.2	2.261E+16	5.00E+00	3.00E+00	7.06E-03	2.16E-01	2.45E-01
50.00	5.719E-01	259.3	1.599E+16	4.95E+00	2.60E+00	3.97E-03	3.14E-01	1.98E-01
55.00	2.990E-01	259.1	8.364E+15	4.85E+00	1.60E+00	2.51E-03	4.84E-01	1.59E-01
60.00	1.550E-01	250.9	4.478E+15	4.50E+00	9.50E-01	1.73E-03	7.15E-01	1.50E-01
65.00	7.900E-02	248.4	2.305E+15	4.00E+00	6.50E-01	1.26E-03	1.07E+00	1.50E-01
70.00	4.000E-02	245.4	1.181E+15	3.30E+00	5.00E-01	9.60E-04	1.52E+00	1.50E-01
75.00	2.000E-02	234.7	6.176E+14	2.70E+00	3.30E-01	7.55E-04	2.17E+00	1.50E-01
80.00	9.660E-03	223.9	3.127E+14	2.00E+00	1.30E-01	6.10E-04	3.06E+00	1.50E-01
85.00	4.500E-03	213.1	1.531E+14	1.33E+00	7.50E-01	5.02E-04	4.56E+00	1.50E-01
90.00	2.022E-03	202.3	7.244E+13	8.50E-01	8.00E-01	4.21E-04	6.88E+00	1.40E-01
95.00	9.070E-04	211.0	3.116E+13	5.40E-01	8.00E-01	3.58E-04	1.06E+01	1.30E-01
100.00	4.230E-04	218.5	1.403E+13	4.00E-01	4.00E-01	3.08E-04	1.71E+01	1.20E-01
105.00	2.070E-04	234.0	6.412E+12	3.40E-01	2.00E-01	2.68E-04	2.47E+01	1.10E-01
110.00	1.080E-04	252.6	3.099E+12	2.80E-01	5.00E-02	2.35E-04	3.36E+01	9.50E-02
115.00	6.000E-05	288.5	1.507E+12	2.40E-01	5.00E-03	2.08E-04	4.15E+01	6.00E-02
120.00	3.590E-05	333.0	7.814E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

Table 1f. Reference Atmospheric Model Profiles, Model 6. U. S. Standard

MODEL = 6 U. S. STANDARD, 1976								
ALT (KM)	PRES (MB)	TEMP (K)	DENSITY (CM-3)	H2O (PPMV)	O3 (PPMV)	N2O (PPMV)	CO (PPMV)	CH4 (PPMV)
0.00	1.013E+03	288.2	2.548E+19	7.75E+03	2.66E-02	3.20E-01	1.50E-01	1.70E+00
1.00	8.988E+02	281.7	2.313E+19	6.07E+03	2.93E-02	3.20E-01	1.45E-01	1.70E+00
2.00	7.950E+02	275.2	2.094E+19	4.63E+03	3.24E-02	3.20E-01	1.40E-01	1.70E+00
3.00	7.012E+02	268.7	1.891E+19	3.18E+03	3.32E-02	3.20E-01	1.35E-01	1.70E+00
4.00	6.166E+02	262.2	1.704E+19	2.16E+03	3.39E-02	3.20E-01	1.31E-01	1.70E+00
5.00	5.405E+02	255.7	1.532E+19	1.40E+03	3.77E-02	3.20E-01	1.30E-01	1.70E+00
6.00	4.722E+02	249.2	1.373E+19	9.25E+02	4.11E-02	3.20E-01	1.29E-01	1.70E+00
7.00	4.111E+02	242.7	1.228E+19	5.72E+02	5.01E-02	3.20E-01	1.25E-01	1.70E+00
8.00	3.565E+02	236.2	1.094E+19	3.67E+02	5.97E-02	3.20E-01	1.19E-01	1.70E+00
9.00	3.080E+02	229.7	9.719E+18	1.58E+02	9.17E-02	3.20E-01	1.09E-01	1.69E+00
10.00	2.650E+02	223.3	8.602E+18	7.00E+01	1.31E-01	3.18E-01	9.96E-02	1.69E+00
11.00	2.270E+02	216.8	7.589E+18	3.61E+01	2.15E-01	3.14E-01	8.96E-02	1.68E+00
12.00	1.940E+02	216.7	6.489E+18	1.91E+01	3.10E-01	3.10E-01	7.81E-02	1.66E+00
13.00	1.658E+02	216.7	5.548E+18	1.09E+01	3.85E-01	3.05E-01	6.37E-02	1.65E+00
14.00	1.417E+02	216.7	4.739E+18	5.93E+00	5.03E-01	3.00E-01	5.03E-02	1.63E+00
15.00	1.211E+02	216.7	4.050E+18	5.00E+00	6.51E-01	2.94E-01	3.94E-02	1.61E+00
16.00	1.035E+02	216.7	3.462E+18	3.95E+00	8.70E-01	2.88E-01	3.07E-02	1.58E+00
17.00	8.850E+01	216.7	2.960E+18	3.85E+00	1.19E+00	2.78E-01	2.49E-02	1.55E+00
18.00	7.565E+01	216.7	2.530E+18	3.83E+00	1.59E+00	2.67E-01	1.97E-02	1.52E+00
19.00	6.467E+01	216.7	2.163E+18	3.85E+00	2.03E+00	2.53E-01	1.55E-02	1.48E+00
20.00	5.529E+01	216.7	1.849E+18	3.90E+00	2.58E+00	2.37E-01	1.33E-02	1.42E+00
21.00	4.729E+01	217.6	1.575E+18	3.98E+00	3.03E+00	2.19E-01	1.23E-02	1.36E+00
22.00	4.047E+01	218.6	1.342E+18	4.07E+00	3.65E+00	2.05E-01	1.23E-02	1.27E+00
23.00	3.467E+01	219.6	1.144E+18	4.20E+00	4.17E+00	1.97E-01	1.31E-02	1.19E+00
24.00	2.972E+01	220.6	9.765E+17	4.30E+00	4.63E+00	1.88E-01	1.40E-02	1.12E+00
25.00	2.549E+01	221.6	8.337E+17	4.43E+00	5.12E+00	1.76E-01	1.50E-02	1.06E+00
27.50	1.743E+01	224.0	5.640E+17	4.58E+00	5.80E+00	1.59E-01	1.60E-02	9.87E-01
30.00	1.197E+01	226.5	3.830E+17	4.73E+00	6.55E+00	1.42E-01	1.71E-02	9.14E-01
32.50	8.010E+00	230.0	2.524E+17	4.83E+00	7.37E+00	1.17E-01	1.85E-02	8.30E-01
35.00	5.746E+00	236.5	1.761E+17	4.90E+00	7.84E+00	9.28E-02	2.01E-02	7.46E-01
37.50	4.150E+00	242.9	1.238E+17	4.95E+00	7.80E+00	6.69E-02	2.22E-02	6.62E-01
40.00	2.871E+00	250.4	8.310E+16	5.03E+00	7.30E+00	4.51E-02	2.50E-02	5.64E-01
42.50	2.060E+00	257.3	5.803E+16	5.15E+00	6.20E+00	2.75E-02	2.82E-02	4.61E-01
45.00	1.491E+00	264.2	4.090E+16	5.23E+00	5.25E+00	1.59E-02	3.24E-02	3.63E-01
47.50	1.090E+00	270.6	2.920E+16	5.25E+00	4.10E+00	9.38E-03	3.72E-02	2.77E-01
50.00	7.978E-01	270.7	2.136E+16	5.23E+00	3.10E+00	4.75E-03*	4.60E-02	2.10E-01
55.00	4.250E-01	260.8	1.181E+16	5.10E+00	1.80E+00	3.00E-03	6.64E-02	1.65E-01
60.00	2.190E-01	247.0	6.426E+15	4.75E+00	1.10E+00	2.07E-03	1.07E-01	1.50E-01
65.00	1.090E-01	233.3	3.386E+15	4.20E+00	7.00E-01	1.51E-03	1.86E-01	1.50E-01
70.00	5.220E-02	219.6	1.723E+15	3.50E+00	3.00E-01	1.15E-03	3.06E-01	1.50E-01
75.00	2.400E-02	208.4	8.347E+14	2.83E+00	2.50E-01	8.89E-04	6.38E-01	1.50E-01
80.00	1.050E-02	198.6	3.832E+14	2.05E+00	3.00E-01	7.06E-04	1.50E+00	1.50E-01
85.00	4.460E-03	188.9	1.711E+14	1.33E+00	5.00E-01	5.72E-04	3.24E+00	1.50E-01
90.00	1.840E-03	186.9	7.136E+13	8.50E-01	7.00E-01	4.71E-04	5.84E+00	1.40E-01
95.00	7.600E-04	188.4	2.924E+13	5.40E-01	7.00E-01	3.93E-04	1.01E+01	1.30E-01
100.00	3.200E-04	195.1	1.189E+13	4.00E-01	4.00E-01	3.32E-04	1.69E+01	1.20E-01
105.00	1.450E-04	208.8	5.033E+12	3.40E-01	2.00E-01	2.84E-04	2.47E+01	1.10E-01
110.00	7.100E-05	240.0	2.144E+12	2.80E-01	5.00E-02	2.44E-04	3.36E+01	9.50E-02
115.00	4.010E-05	300.0	9.688E+11	2.40E-01	5.00E-03	2.12E-04	4.15E+01	6.00E-02
120.00	2.540E-05	360.0	5.114E+11	2.00E-01	5.00E-04	1.85E-04	5.00E+01	3.00E-02

AFGL U.S. STD. PROFILES

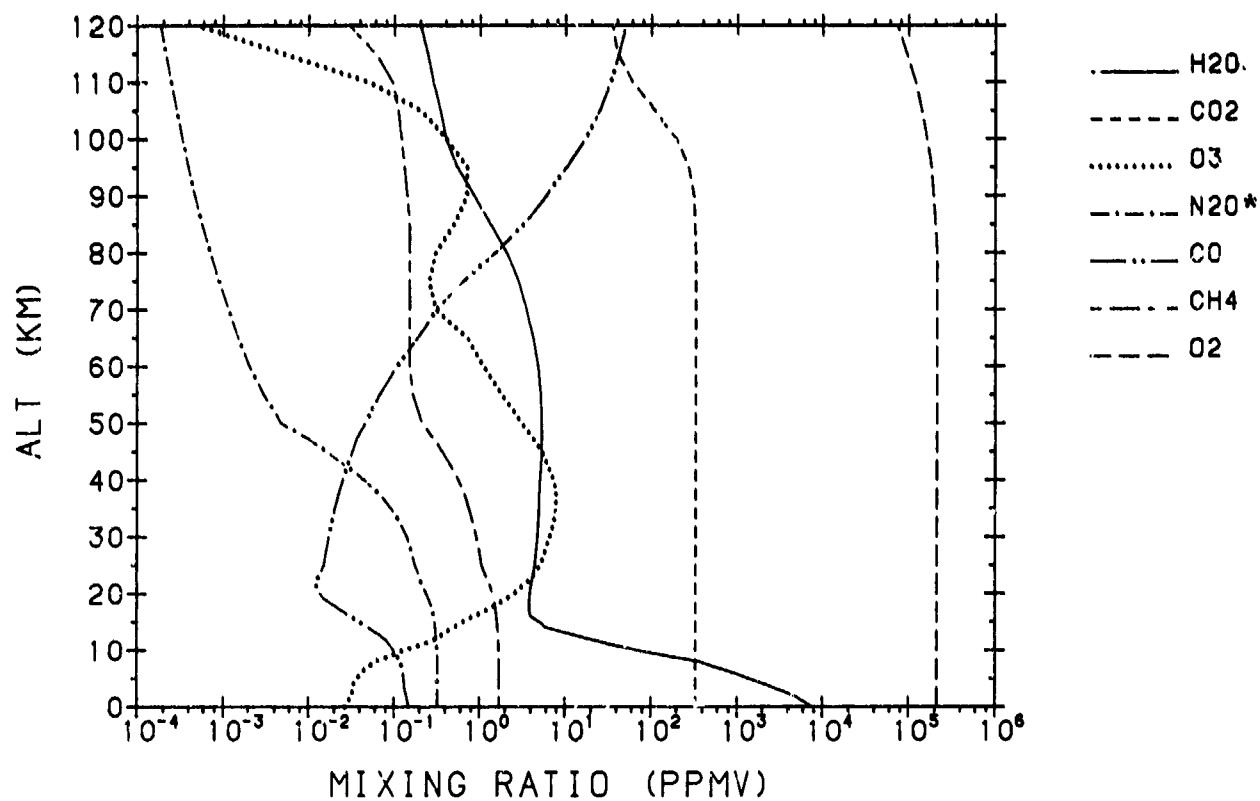


Figure 1. U.S. Standard Model Atmosphere Profiles for the mixing ratios of the major radiating atmospheric gases. This corresponds to Model 6 in Table 1; see text.

[(*) indicates extrapolation adopted for that species; see tabular data]

Table 2. Constituent Profiles (ppmv):

Molecules 1-7 are Model 6 (U.S. Standard) compatible and are repeated here because they duplicate the AFGL Line Parameter and Trace Gas numbering scheme. In addition, CO₂ and O₂ are only defined by a single profile so have not been included in Table 1.

Molecules 8-28 are provided as single representative profiles. Their natural variability can be very different from the selected profile.

[(*) indicates subsequent extrapolation adopted for that species]

Table 2a. Constituent Profiles (ppmv), H₂O, CO₂, O₃, N₂O, CO, CH₄, O₂

	1 H2O	2 CO2	3 O3	4 N2O	5 CO	6 CH4	7 O2
(KM)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)
0.0	7.75E+03	3.30E+02	2.68E-02	3.20E-01	1.50E-01	1.70E+00	2.09E+05
1.0	6.07E+03	3.30E+02	2.93E-02	3.20E-01	1.45E-01	1.70E+00	2.09E+05
2.0	4.03E+03	3.30E+02	3.24E-02	3.20E-01	1.40E-01	1.70E+00	2.09E+05
3.0	3.18E+03	3.30E+02	3.32E-02	3.20E-01	1.35E-01	1.70E+00	2.09E+05
4.0	2.18E+03	3.30E+02	3.39E-02	3.20E-01	1.31E-01	1.70E+00	2.09E+05
5.0	1.40E+03	3.30E+02	3.77E-02	3.20E-01	1.30E-01	1.70E+00	2.09E+05
6.0	9.25E+02	3.30E+02	4.11E-02	3.20E-01	1.29E-01	1.70E+00	2.09E+05
7.0	5.72E+02	3.30E+02	5.01E-02	3.20E-01	1.25E-01	1.70E+00	2.09E+05
8.0	3.67E+02	3.30E+02	5.97E-02	3.20E-01	1.19E-01	1.70E+00	2.09E+05
9.0	1.58E+02	3.30E+02	9.17E-02	3.20E-01	1.09E-01	1.69E+00	2.09E+05
10.0	7.00E+01	3.30E+02	1.31E-01	3.18E-01	9.96E-02	1.69E+00	2.09E+05
11.0	3.61E+01	3.30E+02	2.15E-01	3.14E-01	8.96E-02	1.68E+00	2.09E+05
12.0	1.91E+01	3.30E+02	3.10E-01	3.10E-01	7.81E-02	1.66E+00	2.09E+05
13.0	1.09E+01	3.30E+02	3.85E-01	3.05E-01	6.37E-02	1.65E+00	2.09E+05
14.0	5.93E+00	3.30E+02	5.03E-01	3.00E-01	5.03E-02	1.63E+00	2.09E+05
15.0	5.00E+00	3.30E+02	6.51E-01	2.94E-01	3.94E-02	1.61E+00	2.09E+05
16.0	3.95E+00	3.30E+02	8.70E-01	2.88E-01	3.07E-02	1.58E+00	2.09E+05
17.0	3.85E+00	3.30E+02	1.19E+00	2.78E-01	2.49E-02	1.55E+00	2.09E+05
18.0	3.83E+00	3.30E+02	1.59E+00	2.67E-01	1.97E-02	1.52E+00	2.09E+05
19.0	3.85E+00	3.30E+02	2.03E+00	2.53E-01	1.55E-02	1.48E+00	2.09E+05
20.0	3.90E+00	3.30E+02	2.58E+00	2.37E-01	1.33E-02	1.42E+00	2.09E+05
21.0	3.98E+00	3.30E+02	3.03E+00	2.19E-01	1.23E-02	1.38E+00	2.09E+05
22.0	4.07E+00	3.30E+02	3.65E+00	2.05E-01	1.23E-02	1.27E+00	2.09E+05
23.0	4.20E+00	3.30E+02	4.17E+00	1.97E-01	1.31E-02	1.19E+00	2.09E+05
24.0	4.30E+00	3.30E+02	4.63E+00	1.88E-01	1.40E-02	1.12E+00	2.09E+05
25.0	4.43E+00	3.30E+02	5.12E+00	1.76E-01	1.50E-02	1.06E+00	2.09E+05
27.5	4.58E+00	3.30E+02	5.80E+00	1.59E-01	1.60E-02	9.87E-01	2.09E+05
30.0	4.73E+00	3.30E+02	6.55E+00	1.42E-01	1.71E-02	9.14E-01	2.09E+05
32.5	4.83E+00	3.30E+02	7.37E+00	1.17E-01	1.85E-02	8.30E-01	2.09E+05
35.0	4.90E+00	3.30E+02	7.84E+00	9.28E-02	2.01E-02	7.46E-01	2.09E+05
37.5	4.95E+00	3.30E+02	7.60E+00	6.89E-02	2.22E-02	6.62E-01	2.09E+05
40.0	5.03E+00	3.30E+02	7.30E+00	4.51E-02	2.50E-02	5.64E-01	2.09E+05
42.5	5.15E+00	3.30E+02	6.20E+00	2.75E-02	2.82E-02	4.61E-01	2.09E+05
45.0	5.23E+00	3.30E+02	5.25E+00	1.59E-02	3.24E-02	3.63E-01	2.09E+05
47.5	5.25E+00	3.30E+02	4.10E+00	9.38E-03	3.72E-02	2.77E-01	2.09E+05
50.0	5.23E+00	3.30E+02	3.10E+00	4.75E-03*	4.60E-02	2.10E-01	2.09E+05
55.0	5.10E+00	3.30E+02	1.80E+00	3.00E-03	6.64E-02	1.65E-01	2.09E+05
60.0	4.75E+00	3.30E+02	1.10E+00	2.07E-03	1.07E-01	1.50E-01	2.09E+05
65.0	4.20E+00	3.30E+02	7.00E-01	1.51E-03	1.86E-01	1.50E-01	2.09E+05
70.0	3.50E+00	3.30E+02	3.00E-01	1.15E-03	3.08E-01	1.50E-01	2.09E+05
75.0	2.83E+00	3.30E+02	2.50E-01	8.89E-04	6.38E-01	1.50E-01	2.09E+05
80.0	2.05E+00	3.2E+02	3.00E-01	7.06E-04	1.50E+00	1.50E-01	2.09E+05
85.0	1.33E+00	3.20E+02	5.00E-01	5.72E-04	3.24E+00	1.50E-01	2.00E+05
90.0	8.50E-01	3.10E+02	7.00E-01	4.71E-04	5.84E+00	1.40E-01	1.90E+05
95.0	5.40E-01	2.70E+02	7.00E-01	3.93E-04	1.01E+01	1.30E-01	1.80E+05
100.0	4.00E-01	1.95E+02	4.00E-01	3.32E-04	1.69E+01	1.20E-01	1.60E+05
105.0	3.40E-01	1.10E+02	2.00E-01	2.84E-04	2.47E+01	1.10E-01	1.40E+05
110.0	2.80E-01	6.00E+01	5.00E-02	2.44E-04	3.36E+01	9.50E-02	1.20E+05
115.0	2.40E-01	4.00E+01	5.00E-03	2.12E-04	4.15E+01	6.00E-02	9.40E+04
120.0	2.00E-01	3.50E+01	5.00E-04	1.85E-04	5.00E+01	3.00E-02	7.25E+04

Table 2b. Constituent Profiles (ppmv), NO, SO₂, NO₂, NH₃, HNO₃, OH, HF

	8 NO	9 SO2	10 NO2	11 NH3	12 HNO3	13 OH	14 HF
(KM)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)
0.0	3.00E-04	3.00E-04	2.30E-05	5.00E-04	5.00E-05	4.40E-08	1.00E-08
1.0	3.00E-04	2.74E-04	2.30E-05	5.00E-04	5.96E-05	4.40E-08	1.00E-08
2.0	3.00E-04	2.36E-04	2.30E-05	4.63E-04	6.93E-05	4.40E-08	1.23E-08
3.0	3.00E-04	1.90E-04	2.30E-05	3.80E-04	7.91E-05	4.40E-08	1.97E-08
4.0	3.00E-04	1.46E-04	2.30E-05	2.88E-04	8.87E-05	4.40E-08	3.18E-08
5.0	3.00E-04	1.18E-04	2.30E-05	2.04E-04	9.75E-05	4.40E-08	5.63E-08
6.0	3.00E-04	9.71E-05	2.30E-05	1.46E-04	1.11E-04	4.40E-08	9.18E-08
7.0	3.00E-04	8.30E-05	2.30E-05	9.88E-05	1.26E-04	4.41E-08	1.53E-07
8.0	3.00E-04	7.21E-05	2.30E-05	6.48E-05	1.39E-04	4.45E-08	2.41E-07
9.0	3.00E-04	6.56E-05	2.32E-05	3.77E-05	1.53E-04	4.56E-08	4.04E-07
10.0	3.00E-04	6.08E-05	2.38E-05	2.03E-05	1.74E-04	4.68E-08	6.57E-07
11.0	3.00E-04	5.79E-05	2.62E-05	1.09E-05	2.02E-04	4.80E-08	1.20E-06
12.0	3.00E-04	5.60E-05	3.15E-05	6.30E-06	2.41E-04	4.94E-08	1.96E-06
13.0	2.99E-04	5.59E-05	4.45E-05	3.12E-06	2.76E-04	5.19E-08	3.12E-06
14.0	2.95E-04	5.64E-05	7.48E-05	1.11E-06	3.33E-04	5.65E-08	4.62E-06
15.0	2.83E-04	5.75E-05	1.71E-04	4.47E-07	4.52E-04	6.75E-08	7.09E-06
16.0	2.68E-04	5.75E-05	3.19E-04	2.11E-07	7.37E-04	8.25E-08	1.05E-05
17.0	2.52E-04	5.37E-05	5.19E-04	1.10E-07	1.31E-03	1.04E-07	1.69E-05
18.0	2.40E-04	4.78E-05	7.71E-04	6.70E-08	2.11E-03	1.30E-07	2.57E-05
19.0	2.44E-04	3.97E-05	1.06E-03	3.97E-08	3.17E-03	1.64E-07	4.02E-05
20.0	2.55E-04	3.19E-05	1.39E-03	2.41E-08	4.20E-03	2.16E-07	5.77E-05
21.0	2.77E-04	2.67E-05	1.76E-03	1.92E-08	4.94E-03	3.40E-07	7.77E-05
22.0	3.07E-04	2.28E-05	2.16E-03	1.72E-08	5.46E-03	5.09E-07	9.90E-05
23.0	3.60E-04	2.07E-05	2.58E-03	1.59E-08	5.74E-03	7.59E-07	1.23E-04
24.0	4.51E-04	1.80E-05	3.08E-03	1.44E-08	5.84E-03	1.16E-06	1.50E-04
25.0	6.85E-04	1.75E-05	3.74E-03	1.23E-08	5.61E-03	2.18E-06	1.82E-04
27.5	1.28E-03	1.54E-05	4.81E-03	9.37E-09	4.82E-03	5.00E-06	2.30E-04
30.0	2.45E-03	1.34E-05	6.16E-03	6.35E-09	3.74E-03	1.17E-05	2.83E-04
32.5	4.53E-03	1.21E-05	7.21E-03	3.68E-09	2.59E-03	3.40E-05	3.20E-04
35.0	7.14E-03	1.16E-05	7.28E-03	1.82E-09	1.64E-03	8.35E-05	3.48E-04
37.5	9.34E-03	1.21E-05	6.26E-03	9.26E-10	9.68E-04	1.70E-04	3.72E-04
40.0	1.12E-02	1.36E-05	4.03E-03	2.94E-10*	5.33E-04	2.85E-04	3.95E-04
42.5	1.19E-02	1.65E-05	2.17E-03	8.72E-11	2.52E-04	4.06E-04	4.10E-04
45.0	1.17E-02	2.10E-05	1.15E-03	2.98E-11	1.21E-04	5.11E-04	4.21E-04
47.5	1.10E-02	2.77E-05	6.66E-04	1.30E-11	7.70E-05	5.79E-04	4.24E-04
50.0	1.03E-02	3.56E-05	4.43E-04*	7.13E-12	5.55E-05*	6.75E-04	4.25E-04*
55.0	1.01E-02	4.59E-05	3.39E-04	4.80E-12	4.45E-05	9.53E-04	4.25E-04
60.0	1.01E-02	5.15E-05	2.85E-04	3.66E-12	3.84E-05	1.78E-03	4.25E-04
65.0	1.03E-02	5.11E-05	2.53E-04	3.00E-12	3.49E-05	3.74E-03	4.25E-04
70.0	1.15E-02	4.32E-05	2.31E-04	2.57E-12	3.27E-05	7.19E-03	4.25E-04
75.0	1.61E-02	2.83E-05	2.15E-04	2.27E-12	3.12E-05	1.12E-02	4.25E-04
80.0	2.68E-02	1.33E-05	2.02E-04	2.04E-12	3.01E-05	1.13E-02	4.25E-04
85.0	7.01E-02	5.56E-06	1.92E-04	1.85E-12	2.97E-05	6.10E-03	4.25E-04
90.0	2.13E-01	2.24E-06	1.83E-04	1.71E-12	2.84E-05	1.51E-03	4.25E-04
95.0	7.12E-01	8.98E-07	1.76E-04	1.59E-12	2.78E-05	2.42E-04	4.25E-04
100.0	2.08E+00	3.58E-07	1.70E-04	1.48E-12	2.73E-05	4.47E-05	4.25E-04
105.0	4.50E+00	1.43E-07	1.64E-04	1.40E-12	2.68E-05	1.77E-05	4.25E-04
110.0	7.98E+00	5.73E-08	1.59E-04	1.32E-12	2.64E-05	1.19E-05	4.25E-04
115.0	1.00E+01	2.29E-08	1.55E-04	1.25E-12	2.60E-05	1.35E-05	4.25E-04
120.0	1.00E+01	9.17E-09	1.51E-04	1.19E-12	2.57E-05	2.20E-05	4.25E-04

Table 2c. Constituent Profiles (ppmv), HCl, HBr, HI, ClO, OCS, H₂CO, HOCl

	15 HCL	16 HBR	17 HI	18 CLO	19 OCS	20 H2CO	21 HOCL
(KM)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)
0.0	1.00E-03	1.70E-06	3.00E-06*	1.00E-08	6.00E-04	2.40E-03	7.70E-06
1.0	7.49E-04	1.70E-06	3.00E-06	1.00E-08	5.90E-04	1.07E-03	1.06E-05
2.0	5.61E-04	1.70E-06	3.00E-06	1.00E-08	5.80E-04	4.04E-04	1.22E-05
3.0	4.22E-04	1.70E-06	3.00E-06	1.00E-08	5.70E-04	2.27E-04	1.14E-05
4.0	3.19E-04	1.70E-06	3.00E-06	1.00E-08	5.62E-04	1.40E-04	9.80E-06
5.0	2.39E-04	1.70E-06	3.00E-06	1.00E-08	5.55E-04	1.00E-04	8.01E-06
6.0	1.79E-04	1.70E-06	3.00E-06	1.00E-08	5.48E-04	7.44E-05	6.42E-06
7.0	1.32E-04	1.70E-06	3.00E-06	1.00E-08	5.40E-04	6.04E-05	5.42E-06
8.0	9.96E-05	1.70E-06	3.00E-06	1.01E-08	5.32E-04	5.01E-05	4.70E-06
9.0	7.48E-05	1.70E-06	3.00E-06	1.05E-08	5.25E-04	4.22E-05	4.41E-06
10.0	5.68E-05	1.70E-06	3.00E-06	1.21E-08	5.18E-04	3.63E-05	4.34E-06
11.0	4.59E-05	1.70E-06	3.00E-06	1.87E-08	5.09E-04	3.43E-05	4.65E-06
12.0	4.36E-05	1.70E-06	3.00E-06	3.18E-08	4.98E-04	3.39E-05	5.01E-06
13.0	6.51E-05	1.70E-06	3.00E-06	5.61E-08	4.82E-04	3.50E-05	5.22E-06
14.0	1.01E-04	1.70E-06	3.00E-06	9.99E-08	4.60E-04	3.62E-05	5.60E-06
15.0	1.63E-04	1.70E-06	3.00E-06	1.78E-07	4.26E-04	3.62E-05	6.86E-06
16.0	2.37E-04	1.70E-06	3.00E-06	3.16E-07	3.88E-04	3.58E-05	8.77E-06
17.0	3.13E-04	1.70E-06	3.00E-06	5.65E-07	3.48E-04	3.50E-05	1.20E-05
18.0	3.85E-04	1.70E-06	3.00E-06	1.04E-06	3.09E-04	3.42E-05	1.63E-05
19.0	4.42E-04	1.70E-06	3.00E-06	2.04E-06	2.74E-04	3.39E-05	2.26E-05
20.0	4.89E-04	1.70E-06	3.00E-06	4.64E-06	2.41E-04	3.43E-05	3.07E-05
21.0	5.22E-04	1.70E-06	3.00E-06	8.15E-06	2.14E-04	3.68E-05	4.29E-05
22.0	5.49E-04	1.70E-06	3.00E-06	1.07E-05	1.88E-04	4.03E-05	5.76E-05
23.0	5.75E-04	1.70E-06	3.00E-06	1.52E-05	1.64E-04	4.50E-05	7.65E-05
24.0	6.04E-04	1.70E-06	3.00E-06	2.24E-05	1.37E-04	5.06E-05	9.92E-05
25.0	6.51E-04	1.71E-06	3.00E-06	3.97E-05	1.08E-04	5.82E-05	1.31E-04
27.5	7.51E-04	1.76E-06	3.00E-06	8.48E-05	6.70E-05	7.21E-05	1.84E-04
30.0	9.88E-04	1.90E-06	3.00E-06	1.85E-04	2.96E-05	8.73E-05	2.45E-04
32.5	1.28E-03	2.26E-06	3.00E-06	3.57E-04	1.21E-05	1.01E-04	2.96E-04
35.0	1.57E-03	2.82E-06	3.00E-06	5.08E-04	4.31E-06	1.11E-04	3.21E-04
37.5	1.69E-03	3.69E-06	3.00E-06	6.07E-04	1.60E-06	1.13E-04	3.04E-04
40.0	1.74E-03	4.91E-06	3.00E-06	5.95E-04	6.71E-07	1.03E-04	2.48E-04
42.5	1.76E-03	6.13E-06	3.00E-06	4.33E-04	4.35E-07	7.95E-05	1.64E-04
45.0	1.79E-03	6.85E-06	3.00E-06	2.51E-04	3.34E-07	4.82E-05	9.74E-05
47.5	1.80E-03	7.08E-06	3.00E-06	1.56E-04	2.80E-07	1.63E-05	4.92E-05
50.0	1.80E-03*	7.14E-06*	3.00E-06	1.04E-04*	2.47E-07*	5.10E-06*	2.53E-05*
55.0	1.80E-03	7.15E-06	3.00E-06	7.69E-05	2.28E-07	2.00E-06	1.50E-05
60.0	1.80E-03	7.15E-06	3.00E-06	6.30E-05	2.16E-07	1.05E-06	1.05E-05
65.0	1.80E-03	7.15E-06	3.00E-06	5.52E-05	2.08E-07	6.86E-07	8.34E-06
70.0	1.80E-03	7.15E-06	3.00E-06	5.04E-05	2.03E-07	5.14E-07	7.11E-06
75.0	1.80E-03	7.15E-06	3.00E-06	4.72E-05	1.98E-07	4.16E-07	6.33E-06
80.0	1.80E-03	7.15E-06	3.00E-06	4.49E-05	1.95E-07	3.53E-07	5.78E-06
85.0	1.80E-03	7.15E-06	3.00E-06	4.30E-05	1.92E-07	3.09E-07	5.37E-06
90.0	1.80E-03	7.15E-06	3.00E-06	4.16E-05	1.89E-07	2.76E-07	5.05E-06
95.0	1.80E-03	7.15E-06	3.00E-06	4.03E-05	1.87E-07	2.50E-07	4.78E-06
100.0	1.80E-03	7.15E-06	3.00E-06	3.93E-05	1.85E-07	2.30E-07	4.56E-06
105.0	1.80E-03	7.15E-06	3.00E-06	3.83E-05	1.83E-07	2.13E-07	4.37E-06
110.0	1.80E-03	7.15E-06	3.00E-06	3.75E-05	1.81E-07	1.98E-07	4.21E-06
115.0	1.80E-03	7.15E-06	3.00E-06	3.68E-05	1.80E-07	1.86E-07	4.06E-06
120.0	1.80E-03	7.15E-06	3.00E-06	3.61E-05	1.78E-07	1.75E-07	3.93E-06

Table 2d. Constituent Profiles (ppmv), N₂, HCN, CH₃Cl, H₂O₂, C₂H₂, C₂H₆, PH₃

	22 N2	23 HCN	24 CH3CL	25 H2O2	26 C2H2	27 C2H6	28 PH3
(KM)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)	(PPMV)
0.0	7.81E+05	1.70E-04	7.00E-04	2.00E-04	3.00E-04	2.00E-03	1.00E-14*
1.0	7.81E+05	1.65E-04	6.70E-04	1.95E-04	1.72E-04	2.00E-03	1.00E-14
2.0	7.81E+05	1.63E-04	6.43E-04	1.92E-04	9.57E-05	2.00E-03	1.00E-14
3.0	7.81E+05	1.61E-04	6.22E-04	1.89E-04	6.74E-05	2.00E-03	1.00E-14
4.0	7.81E+05	1.60E-04	6.07E-04	1.84E-04	5.07E-05	1.98E-03	1.00E-14
5.0	7.81E+05	1.60E-04	6.02E-04	1.77E-04	3.99E-05	1.95E-03	1.00E-14
6.0	7.81E+05	1.60E-04	6.00E-04	1.66E-04	3.19E-05	1.90E-03	1.00E-14
7.0	7.81E+05	1.60E-04	6.00E-04	1.49E-04	2.80E-05	1.85E-03	1.00E-14
8.0	7.81E+05	1.60E-04	5.98E-04	1.23E-04	2.55E-05	1.79E-03	1.00E-14
9.0	7.81E+05	1.60E-04	5.94E-04	9.09E-05	2.40E-05	1.72E-03	1.00E-14
10.0	7.81E+05	1.60E-04	5.88E-04	5.79E-05	2.27E-05	1.58E-03	1.00E-14
11.0	7.81E+05	1.60E-04	5.79E-04	3.43E-05	2.08E-05	1.30E-03	1.00E-14
12.0	7.81E+05	1.60E-04	5.66E-04	1.95E-05	1.76E-05	9.86E-04	1.00E-14
13.0	7.81E+05	1.59E-04	5.48E-04	1.08E-05	1.23E-05	7.22E-04	1.00E-14
14.0	7.81E+05	1.57E-04	5.28E-04	6.59E-06	7.32E-06	4.96E-04	1.00E-14
15.0	7.81E+05	1.55E-04	5.03E-04	4.20E-06	4.52E-06	3.35E-04	1.00E-14
16.0	7.81E+05	1.52E-04	4.77E-04	2.94E-06	2.59E-06	2.14E-04	1.00E-14
17.0	7.81E+05	1.49E-04	4.49E-04	2.30E-06	1.55E-06	1.49E-04	1.00E-14
18.0	7.81E+05	1.45E-04	4.21E-04	2.24E-06	8.63E-07	1.05E-04	1.00E-14
19.0	7.81E+05	1.41E-04	3.95E-04	2.68E-06	5.30E-07	7.96E-05	1.00E-14
20.0	7.81E+05	1.37E-04	3.69E-04	3.68E-06	3.10E-07	6.01E-05	1.00E-14
21.0	7.81E+05	1.34E-04	3.43E-04	5.62E-06	1.89E-07	4.57E-05	1.00E-14
22.0	7.81E+05	1.30E-04	3.17E-04	1.03E-05	1.04E-07	3.40E-05	1.00E-14
23.0	7.81E+05	1.25E-04	2.86E-04	1.97E-05	5.75E-08	2.60E-05	1.00E-14
24.0	7.81E+05	1.19E-04	2.48E-04	3.70E-05	2.23E-08	1.89E-05	1.00E-14
25.0	7.81E+05	1.13E-04	1.91E-04	6.20E-05	8.51E-09	1.22E-05	1.00E-14
27.5	7.81E+05	1.05E-04	1.10E-04	1.03E-04	4.09E-09	5.74E-06	1.00E-14
30.0	7.81E+05	9.73E-05	4.72E-05	1.36E-04	2.52E-09	2.14E-06	1.00E-14
32.5	7.81E+05	9.04E-05	1.79E-05	1.36E-04	1.86E-09	8.49E-07	1.00E-14
35.0	7.81E+05	8.46E-05	7.35E-06	1.13E-04	1.52E-09	3.42E-07	1.00E-14
37.5	7.81E+05	8.02E-05	3.03E-06	8.51E-05	1.32E-09	1.34E-07	1.00E-14
40.0	7.81E+05	7.63E-05	1.32E-06	6.37E-05	1.18E-09	5.39E-08*	1.00E-14
42.5	7.81E+05	7.30E-05	8.69E-07	5.17E-05	1.08E-09	2.25E-08	1.00E-14
45.0	7.81E+05	7.00E-05	6.68E-07	4.44E-05	9.97E-10	1.04E-08	1.00E-14
47.5	7.81E+05	6.70E-05	5.60E-07	3.80E-05	9.34E-10	6.57E-09	1.00E-14
50.0	7.81E+05	6.43E-05*	4.94E-07*	3.48E-05	8.83E-10*	4.74E-09	1.00E-14
55.0	7.81E+05	6.21E-05	4.56E-07	3.62E-05	8.43E-10	3.79E-09	1.00E-14
60.0	7.81E+05	6.02E-05	4.32E-07	5.25E-05	8.10E-10	3.28E-09	1.00E-14
65.0	7.81E+05	5.88E-05	4.17E-07	1.26E-04	7.83E-10	2.98E-09	1.00E-14
70.0	7.81E+05	5.75E-05	4.05E-07	3.77E-04	7.60E-10	2.79E-09	1.00E-14
75.0	7.81E+05	5.62E-05	3.96E-07	1.12E-03	7.40E-10	2.66E-09	1.00E-14
80.0	7.81E+05	5.50E-05	3.89E-07	2.00E-03	7.23E-10	2.56E-09	1.00E-14
85.0	7.81E+05	5.37E-05	3.83E-07	1.68E-03	7.07E-10	2.49E-09	1.00E-14
90.0	7.80E+05	5.25E-05	3.78E-07	4.31E-04	6.94E-10	2.43E-09	1.00E-14
95.0	7.79E+05	5.12E-05	3.73E-07	4.98E-05	6.81E-10	2.37E-09	1.00E-14
100.0	7.77E+05	5.00E-05	3.69E-07	6.76E-06	6.70E-10	2.33E-09	1.00E-14
105.0	7.74E+05	4.87E-05	3.66E-07	8.38E-07	6.59E-10	2.29E-09	1.00E-14
110.0	7.70E+05	4.75E-05	3.62E-07	9.56E-08	6.49E-10	2.25E-09	1.00E-14
115.0	7.65E+05	4.62E-05	3.59E-07	1.00E-08	6.40E-10	2.22E-09	1.00E-14
120.0	7.60E+05	4.50E-05	3.56E-07	1.00E-09	6.32E-10	2.19E-09	1.00E-14

The altitude increments for both Table 1 and Table 2 are 1km between 0 and 25km, 2.5km between 25 and 50km, and 5km between 50 and 120km. These increments (and the subset of reference atmospheres themselves) have been chosen for their compatibility with existing profiles in other radiation models (particularly LOWTRAN^{12a, 12b}), facilitating validation and intercomparison tests. The units are: altitude in (km), temperature in (K), pressure in (mb), and mixing ratios in (ppmv). In addition to this document, this profile set is available from AFGL/OPI[‡] in computer-accessible formats, either as tables or FORTRAN data statements appropriate for direct incorporation into computer simulations (e.g. FASCOD2¹³).

A two-part bibliography appears in Appendix B. The first portion is divided by subject: (a) radiance-transmittance models, (b) other constituent compilations and/or photochemical models, (c) reference atmosphere (temperature-pressure) compilations and (d) individual constituents. Each species is followed by the set of journal references which contributed either directly or indirectly to the tabulated profiles. These are generally sequenced according to their influence with secondary sources provided for estimating natural variability and/or uncertainty. For instance, the water vapor profiles incorporate the LOWTRAN¹² values in the tropopause, satellite measurements (LIMS)⁸ in the stratosphere, and photochemical estimates throughout the mesosphere and lower thermosphere^{14, 15}. The particular tabulated values are unique to this compilation, having undergone smoothing, interpolation, and averaging. Similarly, the tropospheric ozone profiles from LOWTRAN have been combined with a composite climatology⁹ based on satellite measurements of the stratosphere (SBUV¹⁶, LIMS¹⁷) and

‡ AFGL/OPI

Hanscom Air Force Base, MA 01731

- 12a. Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Fenn, R.W., and McClatchey, R.A. (1980), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN5, AFGL-TR-80-0067. (NTIS AD A088215).
- 12b. Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Clough, S.A. and Fenn, R.W. (1983), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN6, AFGL-TR-83-0187, (NTIS AD A137796).
13. Clough, S.A., Kneizys, F.X., Shettle, E.P., Anderson, G.P., (1986), Atmospheric radiance and transmittance: FASCOD2, Proc. of the Sixth Conference on Atmospheric Radiation, Williamsburg, VA., American Meteorological Society, Boston, MA., 141-144.
14. Allen, M., Lunine, J.I. and Yung, U.L. (1984) The vertical distribution of ozone in the mesosphere and lower thermosphere, J. Geophys. Res; 89, 4841-4872.
15. Garcia, R.R. and Solomon, S. (1983) A numerical model of the zonally averaged dynamical and chemical structure of the middle atmosphere, J. Geophys. Res; 88, 1379-1400.
16. McPeters, R.D., Heath, D.F. and Bhartia, P.K. (1984) Average ozone profiles for 1979 from the Nimbus 7 SBUV instrument, J. Geophys. Res; 89, 5199-5214.
17. Remsberg, E.E., Russell III, J.M., Gille, J.C., Gordley, L.L., Baily, P.L., Planet, W.G. and Harries, J.E. (1984a) The validation of Nimbus 7 LIMS measurements of ozone, J. Geophys. Res; 89, 5161-5178.

mesosphere (SME¹⁸); however, in the thermosphere all six ozone profiles are identical, based on a single in situ rocket determination (ALADDIN^{19, 20}). The N₂O and CH₄ profiles follow the Smith³ compilation in the tropopause and rely on satellite-derived latitudinal distributions in the stratosphere¹⁰. Above the stratopause the CH₄ profile is photochemically determined^{14, 15} while N₂O has been extrapolated (see subsequent discussion). The second part of the bibliography is an alphabetical listing of all referenced materials; those publications that can be associated with particular molecules are so identified.

3. ERROR ESTIMATES/VARIABILITY

The practical accuracies of these tabulated values vary with species and altitude. At their best they offer approximately 10-30% relative consistency for U.S. Standard Atmosphere conditions throughout the troposphere and stratosphere; exceptions include PH₃ which is unmeasured in the earth's atmosphere. The mesospheric and thermospheric profiles are much less certain and, in fact, are only defined for temperature, pressure, and the following constituents: H₂O, CO₂, O₃, CO, CH₄, O₂, NO, SO₂, OH, and H₂O₂. Mixing ratios for the remaining species have been extrapolated from measurements (usually near the stratopause) using a logarithmically decreasing mixing ratio scale height; the onset of such profile extrapolations is marked by asterisks (*) in the tables and figures. This, of course, leads to unsupported estimates of abundance in the upper atmosphere. [The adopted logarithmic extrapolation scheme is a compromise between using either (a) constant or (b) constantly decreasing mixing ratios. The former introduces erroneous relative changes between extrapolated species. The latter, while obviously connoting the lack of data, introduces an abrupt discontinuity into the profiles.] The mixing ratios of all extrapolated species are, in any case, very small.

4. LIMITATIONS

Representative profiles do not necessarily resemble in situ environments, leading to constraints on their general applicability. WMO and COSPAR will release new sets of

-
18. Thomas, R.J., Barth, C.A. and Solomon, S. (1984) Seasonal variations of O₃ in the upper mesosphere and gravity waves, Geophys. Res. Letts., 11, 673-676.
 19. Weeks, L.H., Good, R.E., Randhawa, J.S. and Trinks, A. (1978) Ozone measurements in the stratosphere, mesosphere, and lower thermosphere during Aladdin 74, J. Geophys. Res. 83, 978-982.
 20. Allen, M., Private Communication.

standard temperature-density profiles in 1986 which provide significant enhancements to the NASA, 1966 Supplements and CIRA, 1972 Reference Atmospheres²¹. (A subset of the CIRA, 1972 profiles is available in this format.) However, a more detailed climatology does not ensure adequate simulation of observed variability. Particularly in disturbed winter conditions, dynamic wave activity can bring about rapid changes in temperature and pressure, which can then propagate from the troposphere into and through the stratosphere.

In addition to any tropospheric meteorologically-driven changes in temperature, the water vapor and anthropogenic pollutants (CO, CO₂, O₃, nitrogen-oxygen compounds, etc) exhibit factors of 100 or more local variability. Dynamic perturbations are less extreme in the stratosphere; however, horizontal gradients on local, latitudinal or seasonal scales often exceed factors of 2-10. In the mesosphere and lower thermosphere, in addition to the extrapolated data, natural excursions brought about by responses to dynamic and solar influences can be substantial. Calculated radiances or transmittances which rely upon default choices represent only a reasonable set of possibilities; they do not replicate actual measurement conditions. When detailed comparisons between theoretical radiance/transmittance calculations and actual data are required, supporting sources (radiosondes, thermosondes, in situ measurements) are recommended.

21. CIRA 1972, (1972) Ed. A.C. Strickland, Akademie-Verlag, Berlin, 450 pp.

References

1. NASA (1966), U.S. Standard Atmosphere Supplements, 1966, U.S. Government Printing Office, Washington, D.C.
2. NASA (1976), U.S. Standard Atmosphere Supplements, 1976, U.S. Government Printing Office, Washington, D.C.
3. Smith, M.A.H. ~~(1982)~~ Compilation of atmospheric gas concentration profiles from 0-50km, NASA Tech Mem 83289, 70 pp.
- 4a. W.M.O. (1982) The Stratosphere 1981: Theory and Measurements, Report No. 11, NASA, Greenbelt, MD.
- 4b. W.M.O. (1986) Atmospheric Ozone 1985: Assessment of Our Understanding of the Processes Controlling its Present Distribution and Change, WMO Report No. 16, W.M.O., Geneva, Switzerland.
5. Brasseur, G. and Solomon, S. (1984) Aeronomy of the Middle Atmosphere, D. Reidel Publishing Co, Dordrecht, Holland, Chapter 5: Composition & Chemistry, 440 pp.
6. Rothman, L.S., Goldman, A., Gillis, J.R., Gamache, R.R., Pickett, H.M., Poynter, R.L., Husson, N. and Chedin, A. (1983b), AFGL trace gas compilation: 1982 version, Appl. Opt., 22, 1616-1627.
7. Rothman, L.S., Gamache, R.R., Barbe, A., Goldman, A., Gillis, H.R., Brown, L.A., Toth, R.A., Flaud, J.-M. and Camy-Peyret, C. (1983a), AFGL atmospheric absorption line parameters compilation: 1982 edition, Appl. Opt., 22, 2247-2256.
8. Russell III, J.M., Gille, J.C., Remsberg, E.E., Gordley, L.L., Bailey, P.L., Fischer, H., Girard, A., Drayon, S.R., Evans, W.F.J. and Harries, J.E. (1984b) Validation of water vapor results measured by the LIMS experiment on Nimbus 7, J. Geophys. Res., 89, 5115-5120.
9. Keating, G.M. and Young D.F. (1985) Interim reference ozone models for the middle atmosphere, Interim COSPAR Reference Atmosphere for Altitudes 20-120km, in preparation.
10. Jones, R.L. and Pyle, J.A. (1984) Observations of CH₄ and N₂O by the Nimbus 7 SAMS: A comparison with in situ data and two-dimensional numerical model calculations, J. Geophys. Res., 89, 5263-5379.

11. Solomon, S., Garcia, R.R., Olivero, J.J., Bevilacqua, R.M., Schwartz, P.R., Clancy, R.T. and Mahleman, D.O. (1985) Photochemistry and transport of CO in the middle atmosphere, J. Atmos. Sci., 42, 1072-1083.
- 12a. Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Fenn, R.W., and McClatchey, R.A. (1980), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN5, AFGL-TR-80-0067, (NTIS AD A088215).
- 12b. Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Clough, S.A. and Fenn, R.W. (1983), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN6, AFGL-TR-83-0187, (NTIS AD A137796).
13. Clough, S.A., Kneizys, F.X., Shettle, E.P., Anderson, G.P., (1986) Atmospheric radiance and transmittance: FASCOD2, Proc. of the Sixth Conference on Atmospheric Radiation, Williamsburg, VA., American Meteorological Society, Boston, MA., 141-144.
14. Allen, M., Lunine, J.I. and Yung, Y.L. (1984) The vertical distribution of ozone in the mesosphere and lower thermosphere, J. Geophys. Res.; 89, 4841-4872.
15. Garcia, R.R. and Solomon, S. (1983) A numerical model of the zonally averaged dynamical and chemical structure of the middle atmosphere, J. Geophys. Res.; 88, 1379-1400.
16. McPeters, R.D., Heath, D.F. and Bhartia, P.K. (1984) Average ozone profiles for 1979 from the Numbus 7 SBUV instrument, J. Geophys. Res.; 89, 5199-5214.
17. Remsberg, E.E., Russell III, J.M., Gille, J.C., Gordley, L.L., Bailey, P.L., Planet, W.G. and Harries, J.E. (1984a) The validation of Nimbus 7 LIMS measurements of ozone, J. Geophys. Res.; 89, 5161-5178.
18. Thomas, R.J., Barth, C.A. and Solomon, S. (1984) Seasonal variations of O₃ in the upper mesosphere and gravity waves, Geophys. Res. Letts., 11, 673-676.
19. Weeks, L.H., Good, R.E., Randhawa, J.S. and Trinks, A. (1978) Ozone measurements in the stratosphere, mesosphere, and lower thermosphere during Aladdin 74, J. Geophys. Res.; 83, 978-982.
20. Allen, M., Private Communication.
21. CIRA 1972, (1972) Ed. A.C. Strickland, Akademie-Verlag, Berlin, 450 pp.

Appendix A

Graphical representation for all tabular data; see the text for description.

Part I: Reference Atmospheric Model Profiles:

	Model No. (1-3)	(4-6)		
Temperature Profiles	(fig. #'s) A1	A2	(Page #'s)	24
Density Profiles	A3	A4		25
Pressure Profiles	A5	A6		26
H ₂ O Mixing Ratios	A7	A8		27
O ₃	A9	A10		28
N ₂ O	A11	A12		29
CO	A13	A14		30
CH ₄	A15	A16		31

Part II: Constituent Profiles (U.S. Standard Atm. ONLY)

H ₂ O, CO ₂ , O ₃ , N ₂ O	(fig. #'s) A17	(Page #'s)	34
CO, CH ₄ , O ₂ , NO	A18		34
SO ₂ , NO ₂ , NH ₃ , HNO ₃	A19		35
OH, HF, HCl, HBr	A20		35
HI, ClO, OCS, H ₂ CO	A21		36
HOCl, N ₂ , HCN, CH ₃ Cl	A22		36
H ₂ O ₂ , C ₂ H ₂ , C ₂ H ₆ , PH ₃	A23		37

Appendix A: Part I

Reference Atmospheric Model Profiles

AFGL TEMPERATURE PROFILES (1-3)

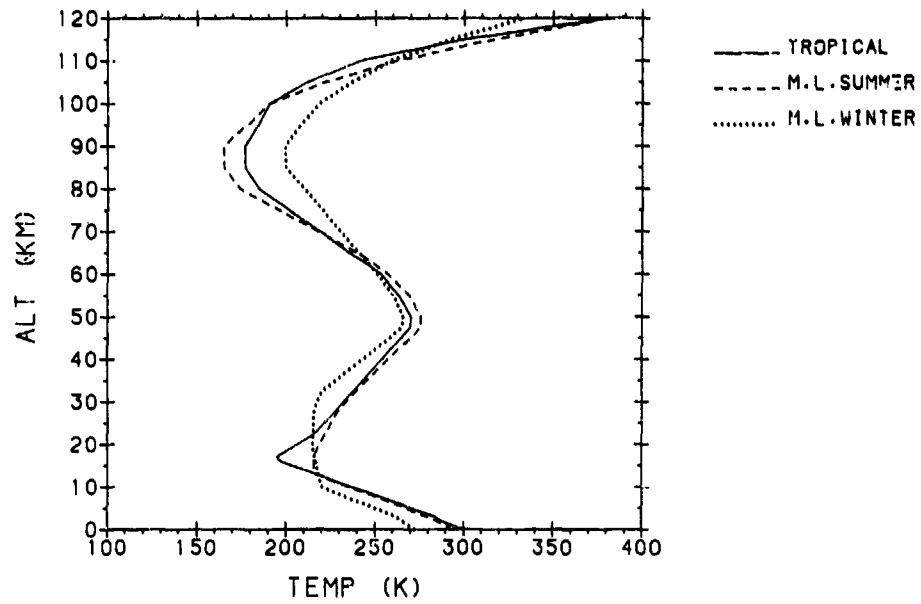


Figure A1. Temperature Profiles (1-3)

AFGL TEMPERATURE PROFILES (4-6)

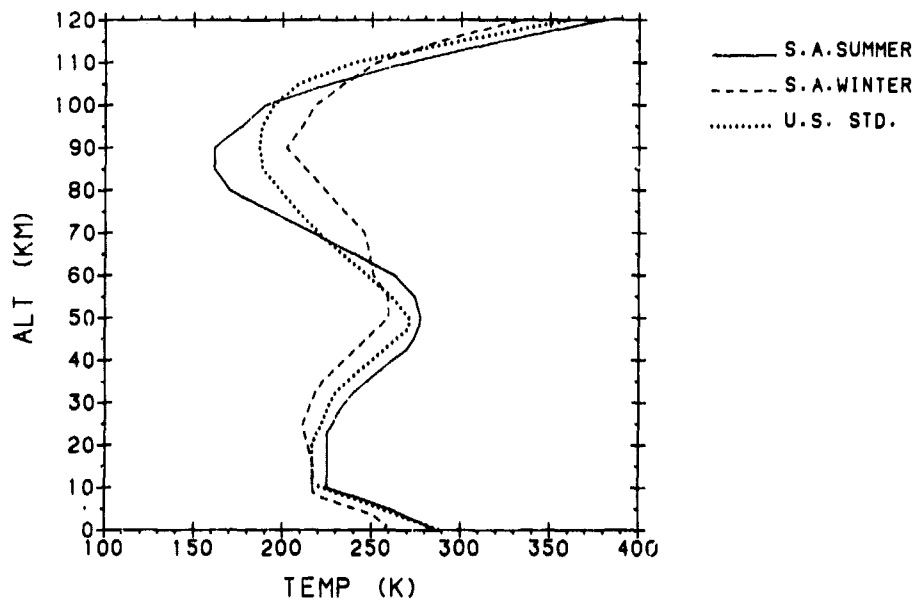


Figure A2. Temperature Profiles (4-6)

AFGL DENSITY PROFILES (1-3)

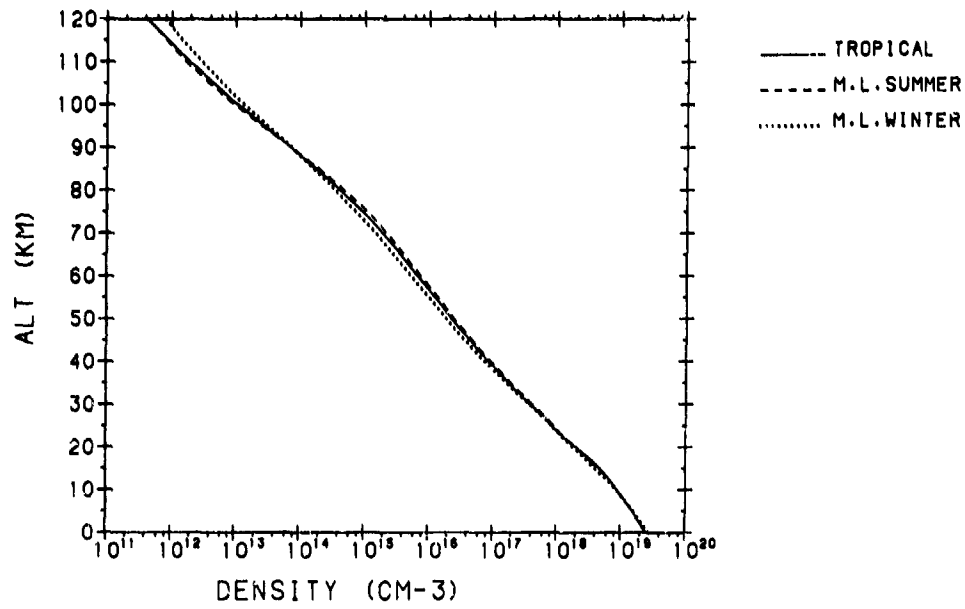


Figure A3. Density Profiles (1-3)

AFGL DENSITY PROFILES (4-6)

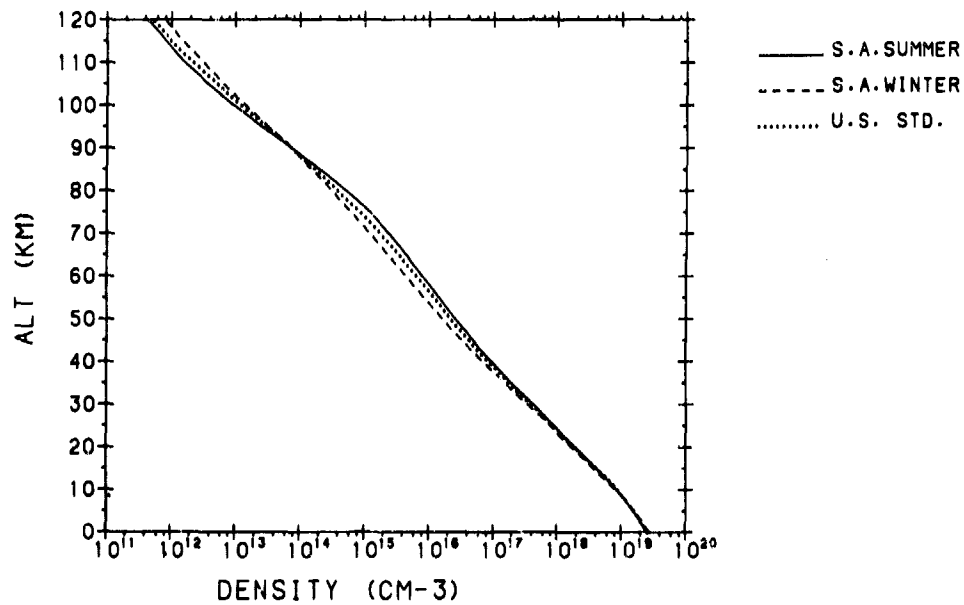


Figure A4. Density Profiles (4-6)

AFGL PRESSURE PROFILES (1-3)

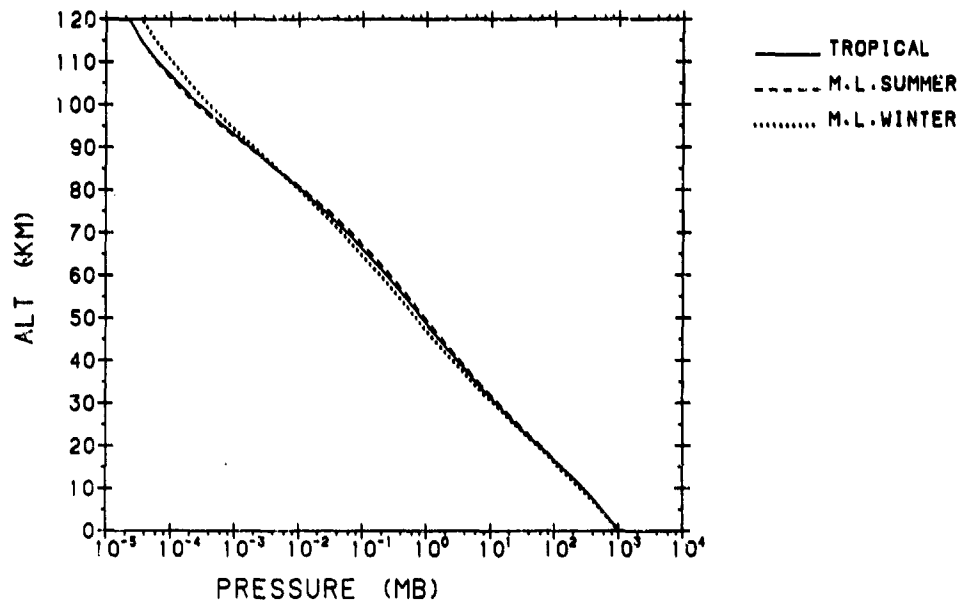


Figure A5. Pressure Profiles (1-3)

AFGL PRESSURE PROFILES (4-6)

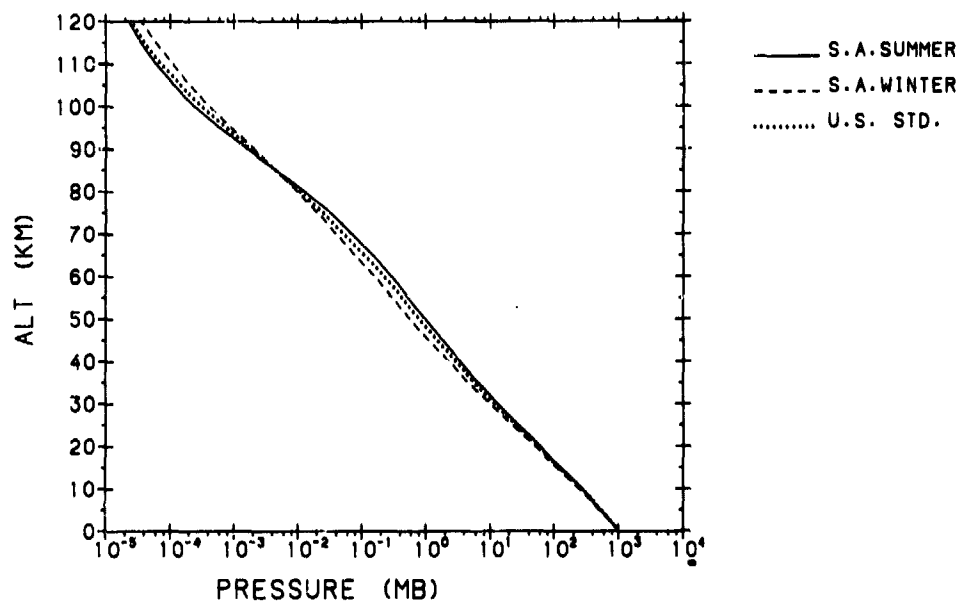


Figure A6. Pressure Profiles (4-6)

AFGL H₂O PROFILES (1-3)

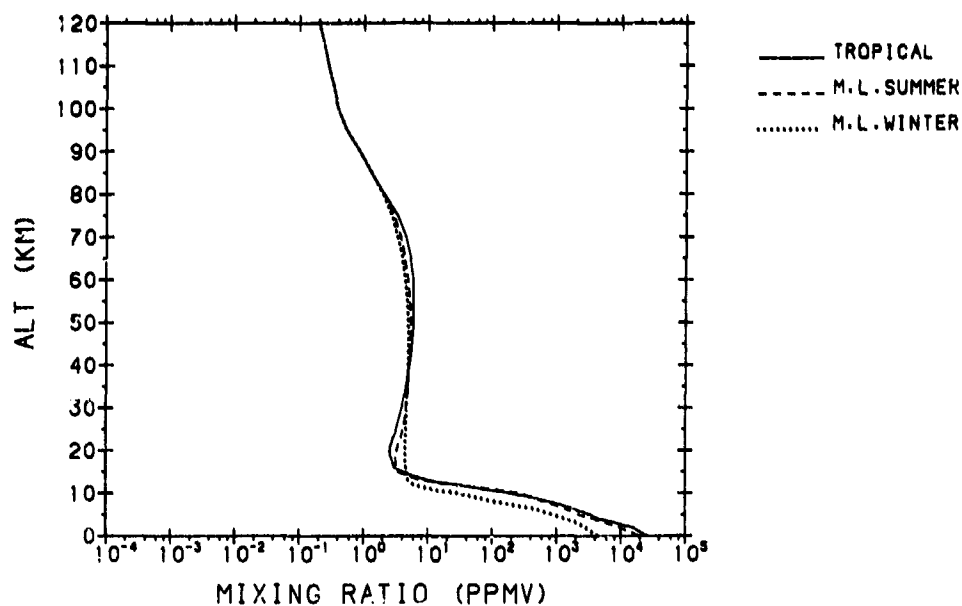


Figure A7. H₂O Mixing Ratios (1-3)

AFGL H₂O PROFILES (4-6)

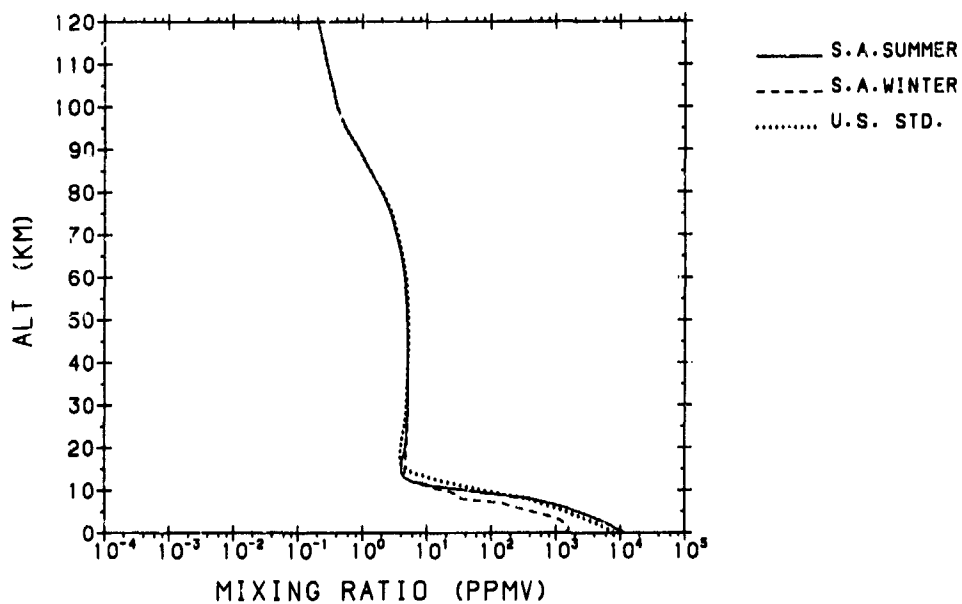


Figure A8. H₂O Mixing Ratios (4-6)

AFGL O3 PROFILES (1-3)

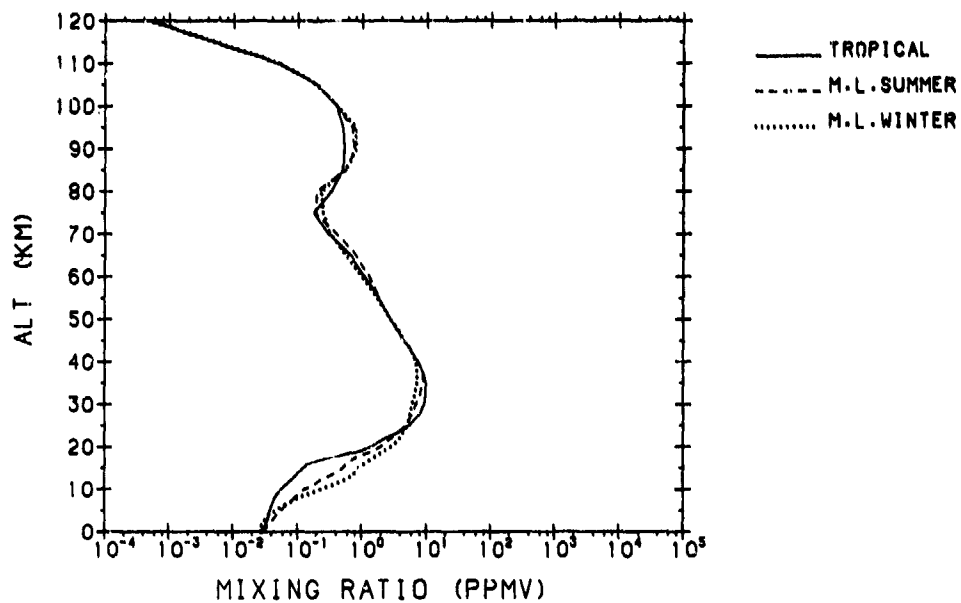


Figure A9. O_3 (1-3)

AFGL O3 PROFILES (4-6)

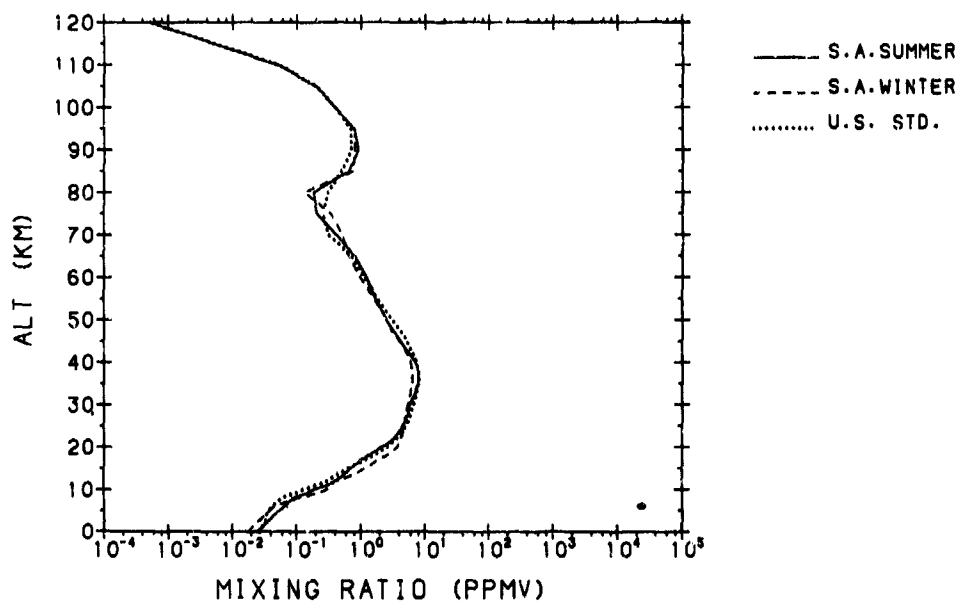


Figure A10. O_3 (4-6)

AFGL N₂O PROFILES (1-3)*

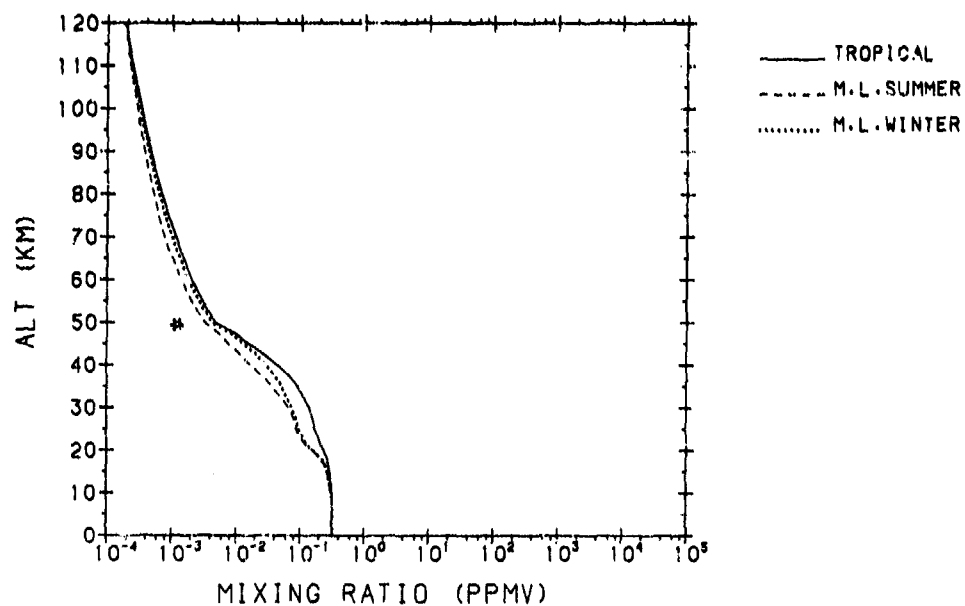


Figure A11. N₂O (1-3)

AFGL N₂O PROFILES (4-6)*

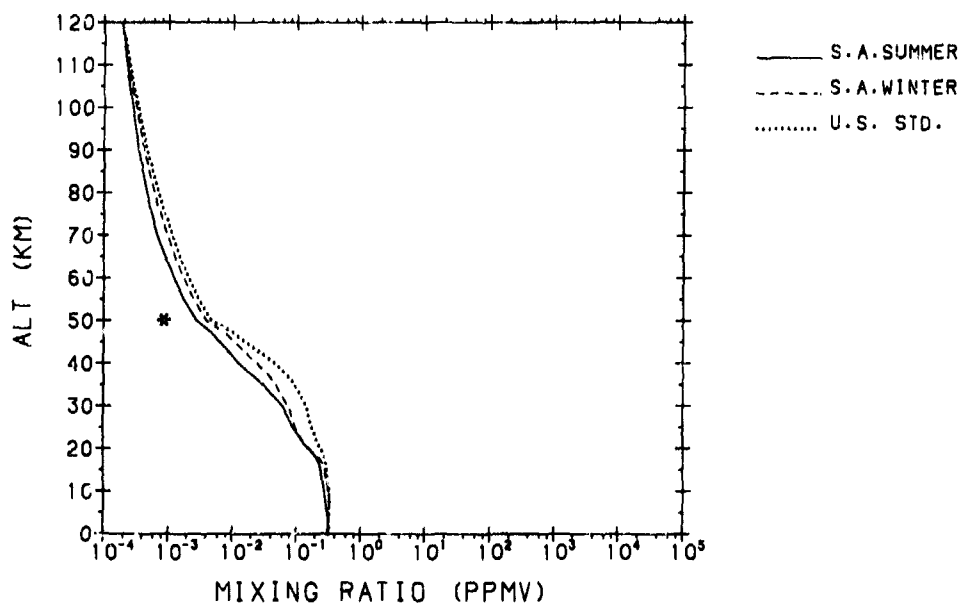


Figure A12. N₂O (4-6)

[(*) indicates extrapolation adopted for that species]

AFGL CO PROFILES (1-3)

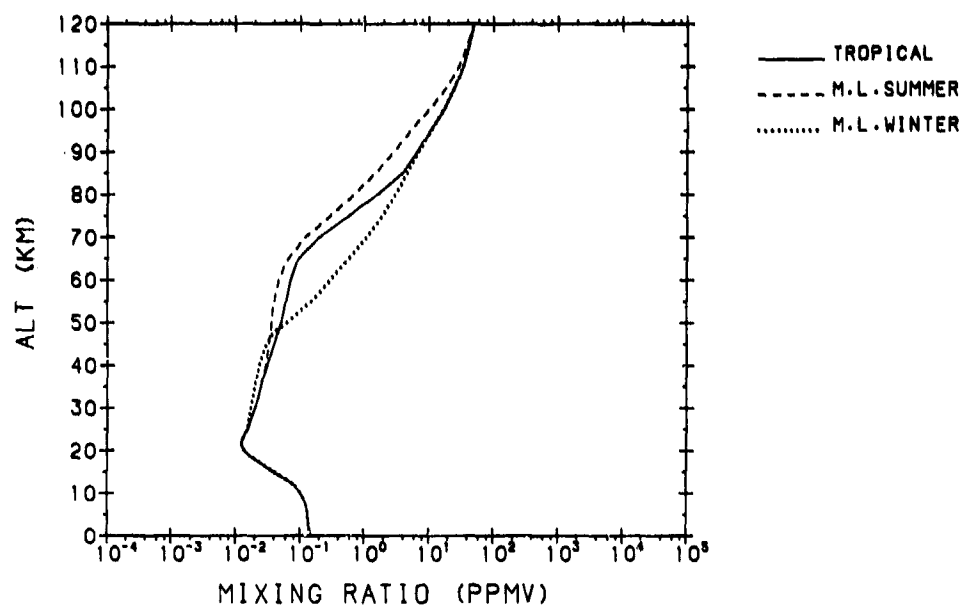


Figure A13. CO (1-3)

AFGL CO PROFILES (4-6)

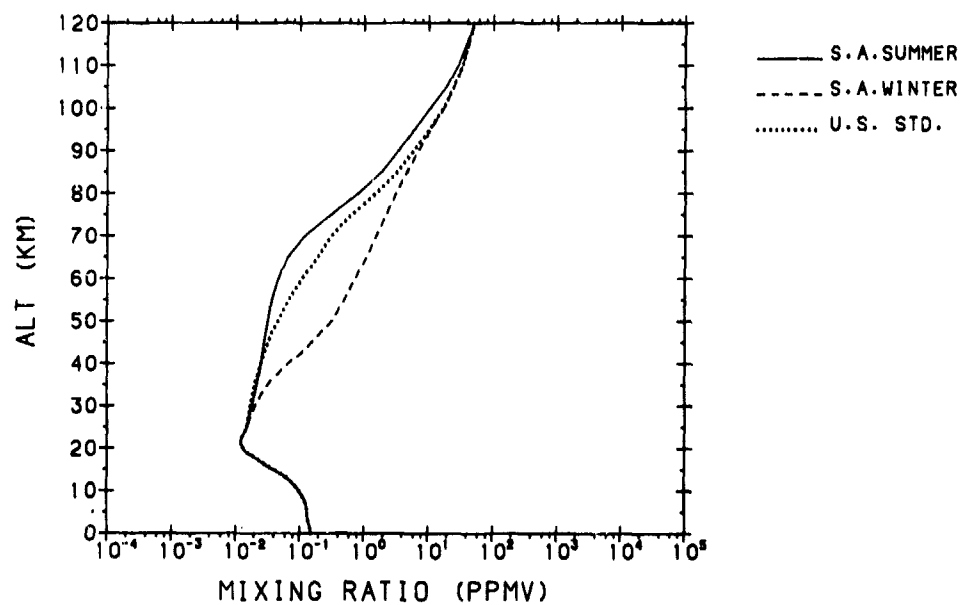


Figure A14. CO (4-6)

AFGL CH₄ PROFILES (1-3)

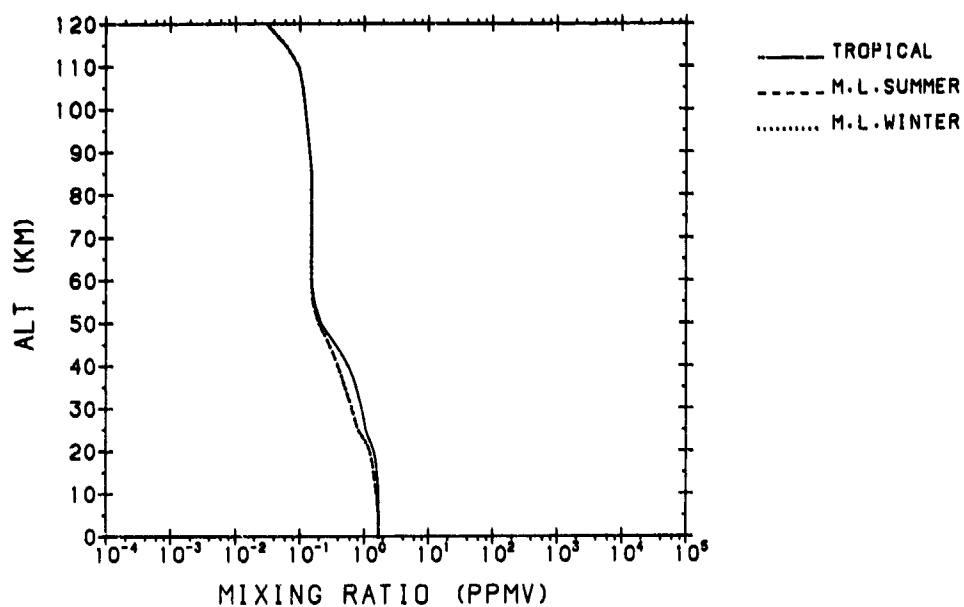


Figure A15. CH₄ (1-3)

AFGL CH₄ PROFILES (4-6)

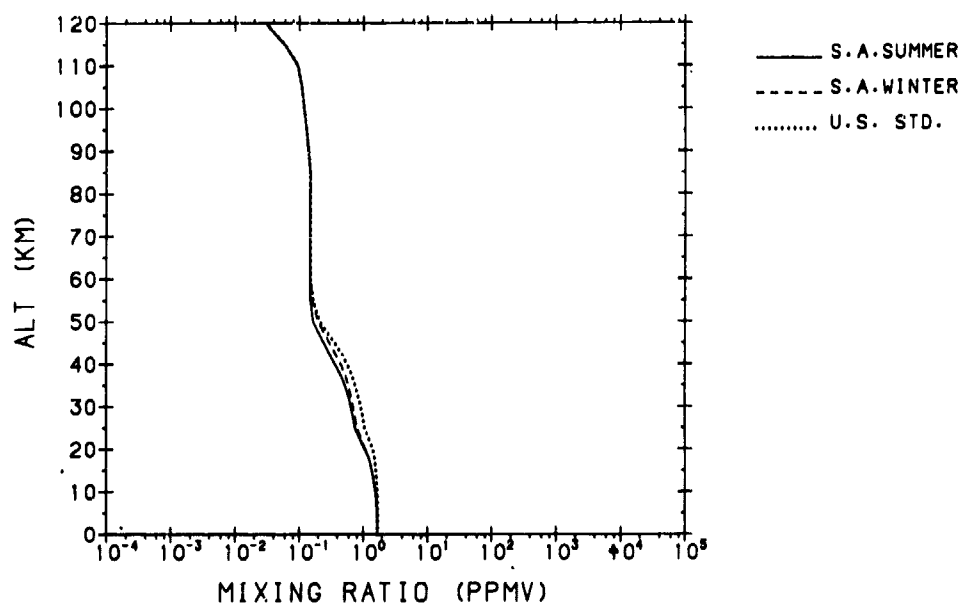


Figure A16. CH₄ (4-6)

Appendix A: Part II

Constituent Profiles (U.S. Standard Atmosphere ONLY)

AFGL CONSTITUENT PROFILES

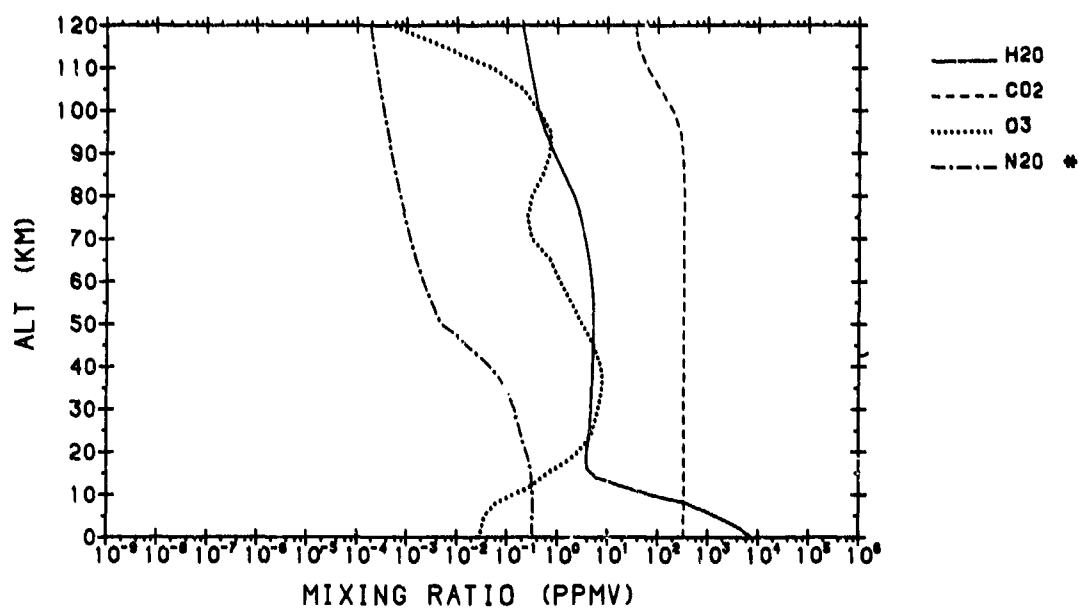


Figure A17. H₂O, CO₂, O₃, N₂O

AFGL CONSTITUENT PROFILES

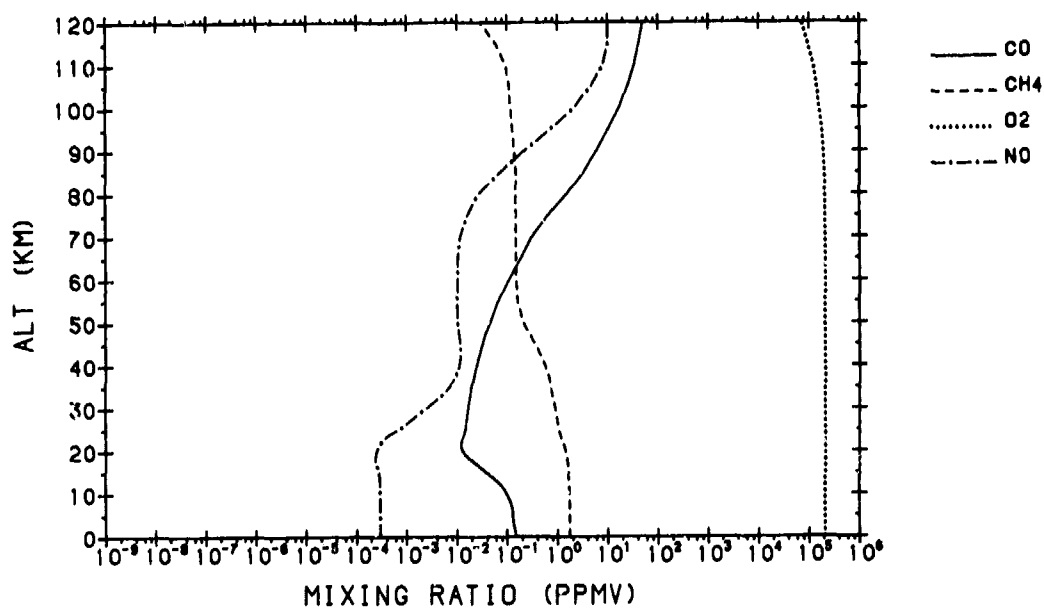


Figure A18. CO, CH₄, O₂, NO

[(*) indicates extrapolation adopted for that species, see tabular data]

AFGL CONSTITUENT PROFILES

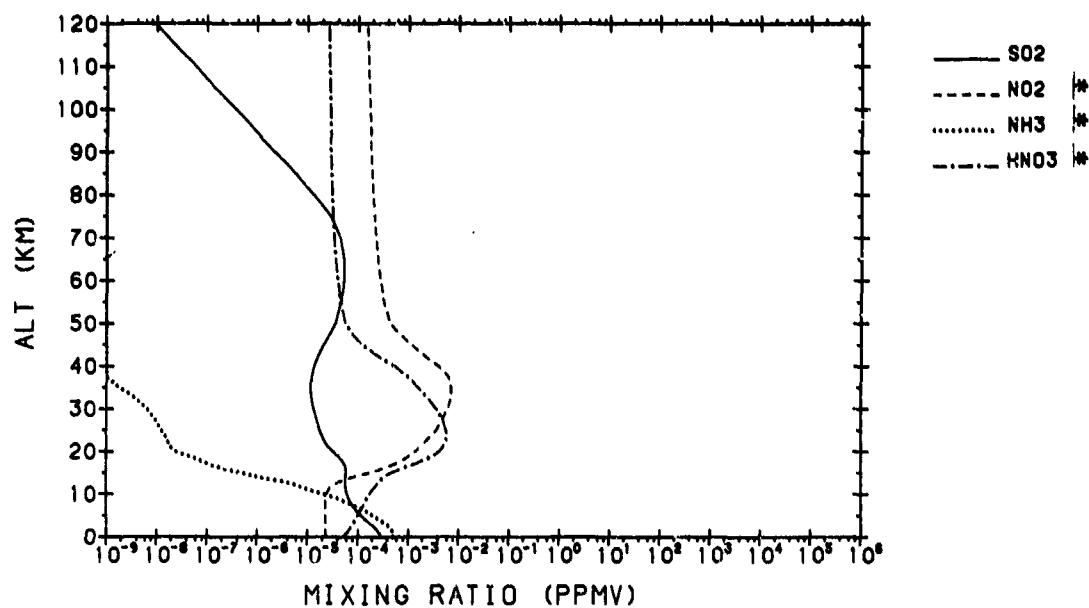


Figure A19. SO₂, NO₂, NH₃, HNO₃

AFGL CONSTITUENT PROFILES

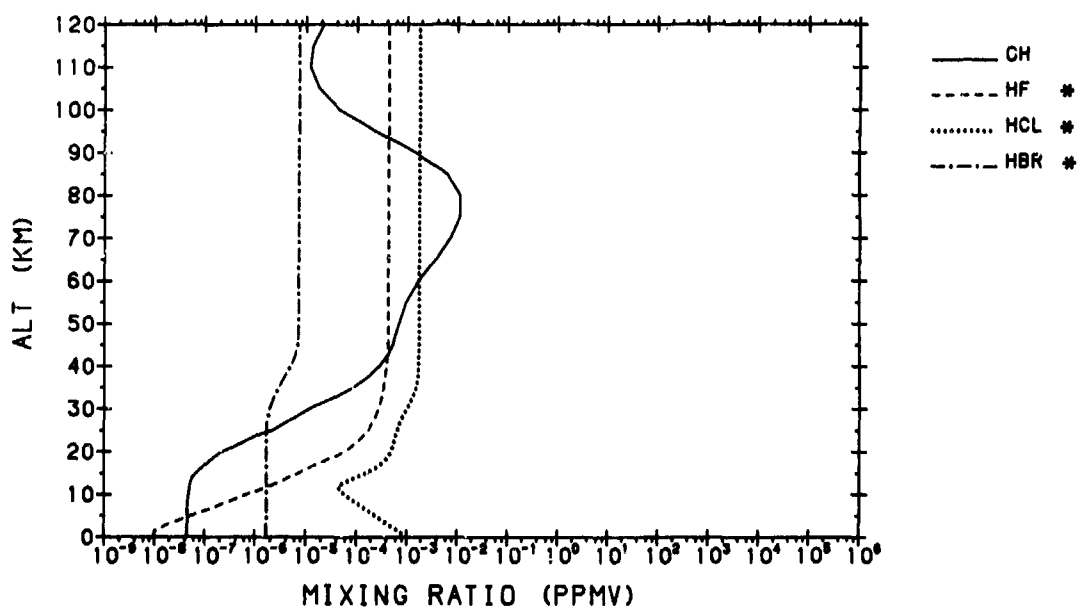


Figure A20. OH, HF, HCl, HBR

[(*) indicates extrapolation adopted for that species, see tabular data]

AFGL CONSTITUENT PROFILES

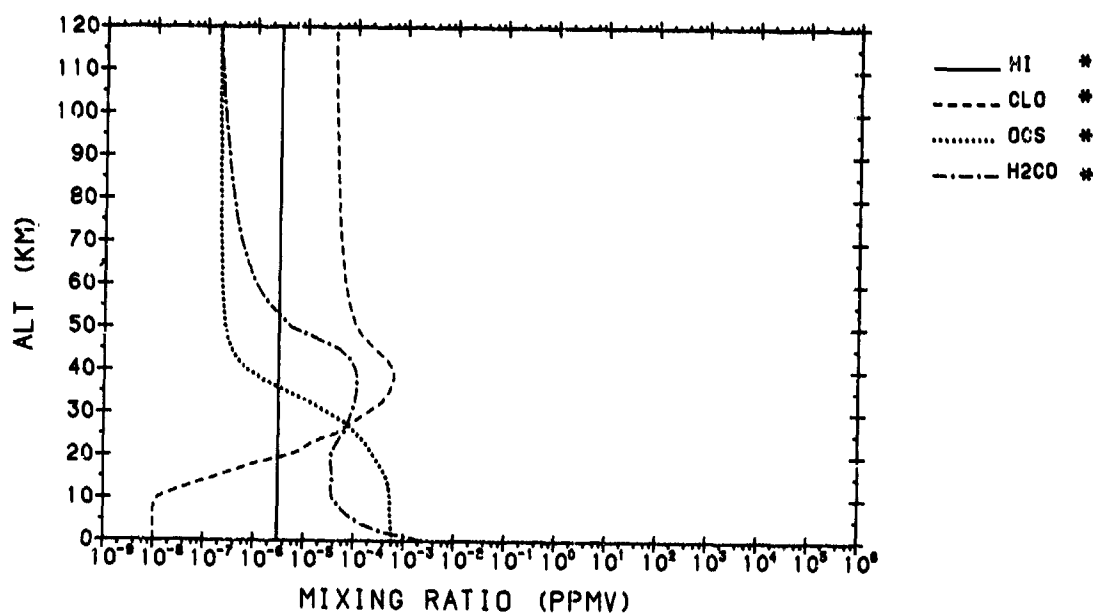


Figure A21. HI, ClO, OCS, H₂CO

AFGL CONSTITUENT PROFILES

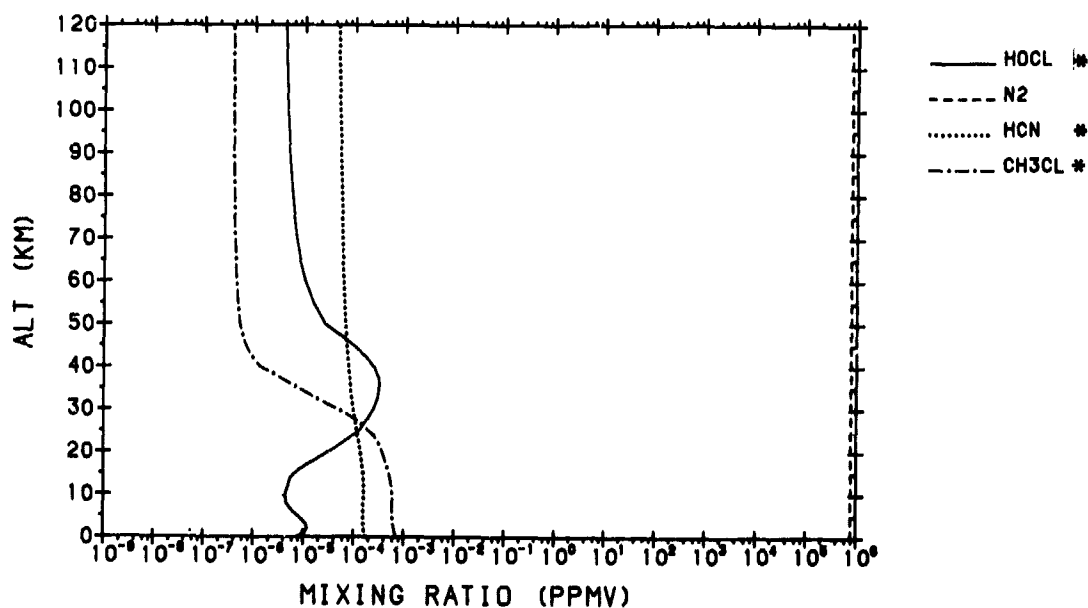


Figure A22. HOCl, N₂, HCN, CH₃Cl

[(*) indicates extrapolation adopted for that species, see tabular data]

AFGL CONSTITUENT PROFILES

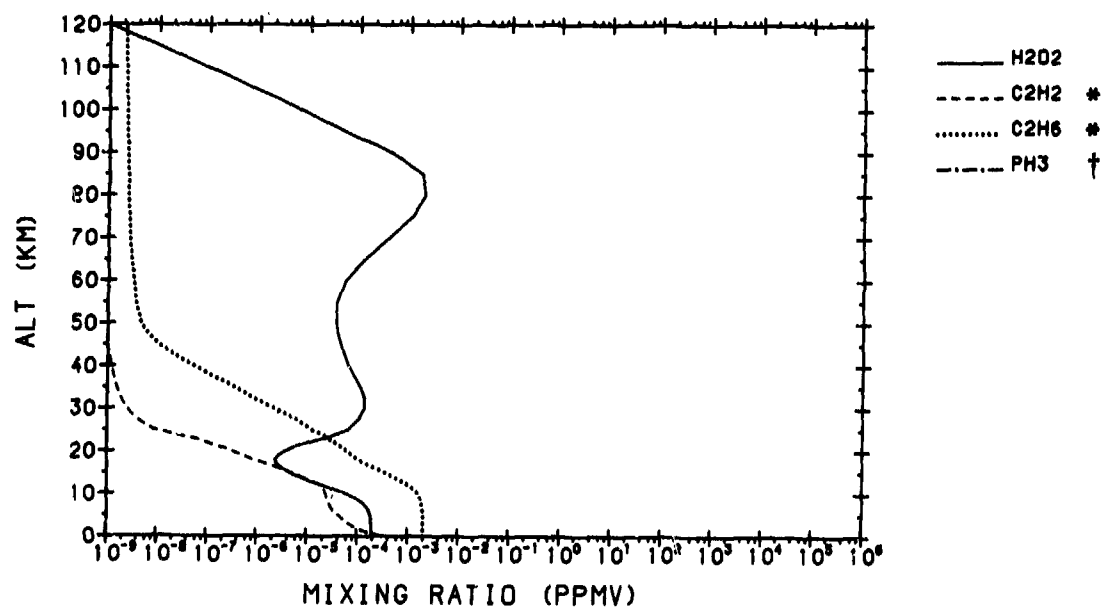


Figure A23. H_2O_2 , C_2H_2 , C_2H_6 , $\text{PH}_3^\dagger$

[(*) indicates extrapolation adopted for that species, see tabular data]

[†] mixing ratio for PH_3 is 1×10^{-14} PPMV]

Appendix B

Bibliography

The tabulated profiles (see text) are, in most cases, derived from more than one literature source. This bibliography provides a list of all references considered when formulating these profiles and is provided as an aide to evaluating species variability, measurement techniques, and/or modeling efforts. In general the final profiles have undergone merging, editing, smoothing, interpolation, and often extrapolation. They can not, therefore, be directly attributed to single sources.

Part I of the Bibliography lists the source data by author and year, separated into four categories: (A) Radiance-Transmittance Models and Line Parameters; (B) Literature Reviews and/or Photochemical Models; (C) Atmospheric Temperature/Pressure Profiles; and (D) Constituent Profiles. The order in which the references appear may reflect their impact upon the adopted profile (see text).

Part II provides the actual alphabetized references with appropriate species identification for cross indexing.

Part I: Subject Listing

A. Radiance-Transmittance Models and Line Parameters

Clough, et al., 1981; Clough, et al., 1985;
Kneizys, et al., 1980; Kneizys, et al., 1983;
Rothman, et al., 1983a; Rothman, et al., 1983b.

B. Literature Reviews/Photochemical Models

Allen, et al., 1984. (subsequently referred to as A)
Brasseur and Solomon, 1984. (referred to as BS)
Crutzen and Gidel, 1983. (referred to as CG)
Garcia and Solomon, 1983. (referred to as GS)
Logan, et al., 1981. (referred to as L)
Smith, 1982. (referred to as Sm)
WMO, 1982. (referred to as WMO)
WMO, 1986.

C. Atmospheric Temperature/Pressure Profiles

NASA, 1966.
CIRA, 1972.
NASA, 1976.

D. Constituent Profiles

[A11]	(Sm); (WMO); (BS)
[H ₂ O]	Kneizys, et al., 1980; Remsberg, et al., 1984b; Farmer, et al., 1980; Girard and Louisnard, 1984; Grossman, et al., 1985; Lippens, et al., 1984; Louisnard, et al., 1983; Russell, et al., 1984b; Stordal, et al., 1985; Weinreb, et al., 1984; Degges and Nadile, in press; Philbrick, private communication; (A); (GS); (L); (Sm); (WMO).
[CO ₂]	Farmer, et al., 1980; Komhyre, et al., 1985; Lippens, et al., 1984; (Sm); (WMO).
[O ₃]	Keating and Young, in press; Bojkov, in press; Fishman, 1985; Girard and Louisnard, 1984; Louisnard, et al., 1983; McPeters, et al., 1984; Remsberg, et al., 1984a; Sellar and Fishman, 1981; (A); (BS); (CG); (GS); (L); (Sm); (WMO).
[N ₂ O]	Jones and Pyle, 1984; Farmer, et al., 1980; Guthrie, et al., 1984; Louisnard, et al., 1983; Rinsland, et al., 1982a; Stordal, et al., 1985; (BS); (Sm); (WMO).
[CO]	Solomon, et al., 1985; Bevilacqua, et al., 1985; Clancy, et al., 1982; Farmer, et al., 1980; Fishman, 1985; Lippens, et al., 1984; Louisnard, et al., 1983; Sellar and Fishman, 1981; Wattenbach, et al., 1984; (A), (BS); (CG); (L); (Sm); (WMO).

[CH ₄]	Jones and Pyle, 1984; Farmer, et al., 1980; Louisnard, et al., 1983; Stordal, et al., 1985; (BS); (CG); (Sm); (WMO).
[O ₂]	NASA, 1976., Allen, private communication; Philbrick, private communication; (A), (BS); (CG); (Sm); (WMO).
[NO]	Grossman, et al., 1985; Hameed and Stewart, 1983; Laurent, et al., 1985; Logan, 1983; Louisnard, et al., 1983; Philbrick, private communication; Rinsland, et al., 1984; (BS); (CG); (L); (Sm); (WMO).
[SO ₂]	Chatfield and Crutzen, 1984; Turco, et al., 1979; (BS); (CG); (Sm); (WMO).
[NO ₂]	Hameed and Stewart, 1983; Laurent, et al., 1985; Logan, 1983; Louisnard, et al., 1993; Naudet, et al., 1980; Russell, et al., 1984a; Russell, et al., 1984c; Solomon, et al., 1984; Weis, et al., 1984; (BS); (CG); (L); (Sm); (WMO).
[NH ₃]	Arijs, et al., 1982; Farmer and Dawson, 1982; Oelhaf, et al., 1983; (BS); (Sm); (WMO).
[HNO ₃]	Gille, et al., 1984; Girard and Louisnard, 1984; Goldman, et al., 1984; Logan, 1983; Weinreb, et al., 1984; (BS); (Sm); (WMO).
[OH]	Chatfield and Crutzen, 1984; Hameed and Stewart, 1983; Herman and McQuillan, 1985; Philbrick, private communication; Solomon, et al., 1982; (BS); (CG); (L); (Sm); (WMO).
[HF]	Farmer, et al., 1980; Mankin and Coffey, 1983; (BS); (Sm); (WMO).
[HCl]	Farmer, et al., 1980; Herman and McQuillan, 1985; Mankin and Coffey, 1983.
[HBr]	(BS); (Sm); (WMO).
[HI]	(BS); (Sm); (WMO).
[ClO]	Herman and McQuillan, 1985; Weinstock, et al., 1981; (BS); (Sm); (WMO).
[OCS]	Louisnard, et al., 1983; Turco, et al., 1979; (BS); (Sm); (WMO).
[H ₂ CO]	(BS); (Sm); (WMO).
[HOCl]	Herman and McQuillan, 1985; (BS); (Sm); (WMO).
[N ₂]	(BS); (Sm); (WMO).
[HCN]	Rinsland, et al., 1982; Smith and Rinsland, 1985; (BS); (Sm); (WMO).
[CH ₂ Cl]	(CG), (L), (Sm), (WMO).
[H ₂ O ₂]	Chatfield and Crutzen, 1984; (BS); (Sm); (WMO).
[C ₂ H ₂]	Brewer, et al., 1983; Goldman, et al., 1981; Rasmussen, et al., 1983; (BS); (Sm); (WMO).
[C ₂ H ₆]	Rasmussen, et al., 1983; (BS); (L); (Sm); (WMO).
[PH ₃]	(Sm).

Part II: Author Listing

- Abbas, M.M., Guo, J., Nolt, J.G. (1984) Far infrared remote sounding of stratospheric temperature and trace gas distribution, J. Atmos. Chem., 2, 145-162. [OH, HCl].
- Allen, M., Lunine, J.I. and Yung, Y.L. (1984) The vertical distribution of ozone in the mesosphere and lower thermosphere, J. Geophys. Res., 89, 4841-4872. [O₃, H₂O₂, H₂O]
- Allen, M., Yung, Y.L. and Waters, J.W. (1981) Vertical transport and photochemistry in the terrestrial mesosphere and lower thermosphere (50-120km), J. Geophys. Res., 86, 3617-3627. [CO]
- Arijs, E., Nevejans, D. and Ingles, J. (1982) Stratospheric positive ion composition measurements, ion abundances and related trace gas detection, J. Atmos. Terrestrial Phys., 44, 45-53. [NH₃]
- Bevilacqua, R.M., Stark, A.A. and Schwartz, P.R. (1985) The variability of carbon monoxide in the terrestrial mesosphere as determined from ground-based observations of the J=1→0 emission line, J. Geophys. Res., 90, 577-5782. [CO]
- Bojkov, R.D. (in press), Tropospheric Ozone, its changes and possible radiative effects, WMO Special Environmental Report #16. [O₃]
- Brasseur, G. and Solomon, S. (1984) Aeronomy of the Middle Atmosphere, D. Reidel Publishing CO, Dordrecht, Holland, Chapter 5: Composition & Chemistry, 440 pp. [ClO, HCl, HOCl, others]
- Brewer, D.A., Augustsson, T.R. and Levine, J.S. (1983) The photochemistry of anthropogenic nonmethane hydrocarbons in the troposphere, J. Geophys. Res., 88, 6683-6695. [C₂H₂]
- Chatfield, R.B. and Crutzen, P.J. (1984) Sulfur dioxide in remote oceanic air: cloud transport of reactive precursors, J. Geophys. Res., 89, 7111-7132. [OH, H₂O₂, SO₂]
- Cicerone, R.J. and Zellner, R. (1983) The atmospheric chemistry of hydrogen cyanide (HCN), J. Geophys. Res., 88, 10689-10696. [HCN]
- CIRA 1972, (1972) Ed. A.C. Strickland, Akademie-Verlag, Berlin, 450 pp.
- Clancy, R.T., Muhleman, D.O. and Berge, G.L. (1982) Microwave spectra of terrestrial mesospheric CO, J. Geophys. Res., 87, 5009-5014. [CO]
- Clough, S.A., Kneizys, F.X., Rothman, L.S. and Gallery, W.O. (1981), Atmospheric spectral transmittance and radiance: FASCOD1B, SPIE, 277, Pg. 152-166.
- Clough, S.A., Kneizys, F.X., Shettle, E.P., Anderson, G.P., (1986) Atmospheric radiance and transmittance: FASCOD2, Proc. of the Sixth Conference on Atmospheric Radiation, Williamsburg, VA., American Meteorological Society, Boston, MA., 141-144.
- Crutzen, P.J. and Gidel, L.T. (1983) A two-dimensional photochemical model of the atmosphere 2: The tropospheric budgets of the anthropogenic chlorocarbons CO, CH₄, CH₃Cl and the effects of various NO_x sources on tropospheric ozone, J. Geophys. Res., 88, 6641-6661. [NO, CO, CH₄, CH₃Cl, O₃, HNO₃, OH, N₂O₅]
- Degges, T.C. and Nadile, R. (in preparation) Determination of mesospheric water vapor concentrations from earth limb radiance measurements at 6.3 and 2.7 micrometers. [H₂O]
- Ehhalt, D.H., Roth, E.P., Schmidt, U. (1983) On the temporal variance of stratospheric trace gas concentrations, J. Atmos. Chem., 1, 27-51. [N₂O, CH₄]
- Farmer, J.C. and Dawson, G.A. (1982) Condensation sampling of soluble atmospheric trace gases, J. Geophys. Res., 87, 8931-8942. [NH₃]

- Farmer, C.B., Raper, O.F., Robbins, B.D., Toth, R.A. and Muller, C. (1980) Simultaneous spectroscopic measurements of stratospheric species: O_3 , CH_4 , CO , CO_2 , N_2O , HCl and HF at northern and southern mid latitudes, J. Geophys. Res.; 85, 1621-1632. [H_2O , HCl , HF , O_3 , CH_4 , CO , CO_2 , N_2O]
- Fishman, J. (1985) Ozone in the troposphere, in Ozone in the Free Atmosphere, ed R.C. Whitten & S. Prasad, Von Nostrand Reinhold Pub. [CO , O_3]
- Gallery, W.O., Kneizys, F.X. and Clough, S.A. (1983), Air Mass Transmittance/Radiance Calculation: FASCAT, AFGL-TR-83-0065, (NTIS AD A132108).
- Garcia, R.R. and Solomon, S. (1983) A numerical model of the zonally averaged dynamical and chemical structure of the middle atmosphere, J. Geophys. Res.; 88, 1379-1400. [HO_2 , NO_2 , H_2O_2 , NO , HNO_3 , OH , CH_4 , H_2O , H_2]
- Gille, J.C., Russell III, J.M., Bailey, P.L., Remsberg, E.E., Gordley, L.L., Evans, W.F.J., Fischer, H., Gandrud, B.W., Girard, A., Harries, J.E. and Beck, S.A. (1984) Accuracy & precision of the nitric acid concentrations determined by the LIMS experiment on Nimbus 7, J. Geophys. Res.; 89, 5179-5190. [HNO_3]
- Girard, A. and Louisnard, N. (1984) Stratospheric water vapor, nitrogen dioxide, nitric acid, and ozone measurements deduced from spectroscopic observations, J. Geophys. Res.; 89, 5109-5114. [H_2O , NO_2 , HNO_3 , O_3]
- Goldman, A., Murcray, F.J., Blatherwick, R.D., Gillis, J.R., Bonomo, F.S., Murcray, F.H. and Murcray, D.G. (1981) Identification of acetylene (C_2H_2) in infrared atmospheric absorption spectra, J. Geophys. Res.; 86, 12143-12146. [C_2H_2]
- Goldman, A., Gillis, J.R., Rinsland, C.P., Murcray, F.J. and Murcray, D.G. (1984) Stratospheric HNO_3 quantification from line-by-line nonlinear least-squares analysis of high resolution balloon-borne solar absorption spectra in the 870 cm^{-1} region, Applied Optics, 23, 3252-3255. [HNO_3]
- Goldman, A., Rinsland, C.P., Murcray, F.J., Murcray, D.G., Coffey, M.T., Mankin, W.A. (1984) Balloon-borne and aircraft infrared measurements of ethane (C_2H_6) in the upper troposphere and lower stratosphere, J. Atmos. Chem. (2), 211-221. [C_2H_6]
- Grossman, K.U., Frings, W.G., Offermann, D., Andre, L., Kopp, E. and Krankowsky, D. (1985) Concentrations of H_2O and NO in the mesosphere and the lower thermosphere at high latitudes, J. Atmos. Terr. Phys.; 47, 291-300. [H_2O , NO]
- Guthrie, P.D., Jackman, C.H., Herman, J.R. and McQuillan, C.J. (1984) A diabatic circulation experiment in a two-dimensional photochemical model, J. Geophys. Res.; 89, 9589-9602. [N_2O]
- Hameed, S. and Stewart, R.W. (1983) Latitudinal variation of tropospheric ozone in a photochemical model, J. Geophys. Res.; 88, 5153-5162. [OH , NO , NO_2]
- Herman, J.R. and McQuillan, C.J. (1985) Atmospheric chlorine & stratospheric ozone nonlinearities and trend detections, J. Geophys. Res.; 90, 5721-5732. [ClO , $HOCl$, OH , HCl]
- Jones, R.L. and Pyle, J.A. (1984) Observations of CH_4 and N_2O by the Nimbus 7 SAMS: A comparison with in situ data and two-dimensional numerical model calculations, J. Geophys. Res.; 89, 5263-5379. [CH_4 , N_2O]
- Keating, G.M. and Young D.F. (1985) Interim reference ozone models for the middle atmosphere, Interim COSPAR Reference Atmosphere for Altitudes 20-120km, in preparation. [O_3]
- Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Fenn, R.W., and McClatchey, R.A. (1980), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN5, AFGL-TR-80-0067, (NTIS AD A088215).

- Kneizys, F.X., Shettle, E.P., Gallery, W.O., Chetwynd, J.H., Abreu, L.W., Selby, J.E.A., Clough, S.A. and Fenn, R.W. (1983), Atmospheric Transmittance/Radiance: Computer Code LOWTRAN6, AFGL-TR-83-0187, (NTIS AD A137796).
- Komhyre, W.D., Gammon, R.H., Harris, T.D., Waterman, L.S., Conway, T.J., Taylor, W.R. and Thoning (1985) Global atmospheric CO₂ distribution and variations from 1988-1982 NOAA/GMCC CO₂ flask sample data, J. Geophys. Res., 90, 5567-5596. [CO₂]
- Laurent, J., Lemaitre, M.P., Besson, J., Girard, A., Lippens, C., Muller, C., Vercheval, J. and Ackerman, M. (1985) Middle atmospheric NO and NO₂ observed by the spacelab grille spectrometer, Nature, 315, 126-127. [NO, NO₂]
- Lippens, C., Muller, C., Vercheval, J., Ackerman, M., Laurent, J., Lemaitre, M., Besson, J. and Girard, A. (1984) Trace constituents measurements deduced from spectrometric observations on-board spacelab, Adv. Space Res., 4, 75-79. [CO, CO₂, H₂O]
- Logan, J.A. Prather, M.J. Wofsy, S.A. and McElroy, M.B. (1981) Tropospheric chemistry: A global perspective, J. Geophys. Res.; 86, 7210-7254. [H₂O₂, CH₃Cl, C₂H₆, NO, NO₂, OH, HO₂, H₂O, O₃, CO, HNO₃]
- Logan, J.A. (1983) Nitrogen oxides in the troposphere: global & regional budgets, J. Geophys. Res.; 88, 10785-10807. [NO, NO₂, HNO₃, NO₃]
- Louisnard, N., Fergant, G., Girard, A., Gramont, L., Lado-Bordowsky, O., Laurent, J., LeBoiteux, S. and Lemaitre, M.P. (1983) Infrared absorption spectroscopy applied to stratospheric profiles of minor constituents, J. Geophys. Res.; 88, 5365-5376. [CO, COS, CH₄, H₂O, O₃, NO, NO₂, N₂O, HNO₃]
- Mankin, W.G. and Coffey, M.T. (1983) Latitudinal distributions and temporal changes of stratospheric HCl and HF, J. Geophys. Res.; 88, 10776-10784. [HCl, HF]
- McPeters, R.D., Heath, D.F. and Bhartia, P.K. (1984) Average ozone profiles for 1979 from the Nimbus 7 SBUV instrument, J. Geophys. Res.; 89, 5199-5214. [O₃]
- NASA (1966), U.S. Standard Atmosphere Supplements, 1966, U.S. Government Printing Office, Washington, DC.
- NASA (1976), U.S. Standard Atmosphere Supplements, 1976, U.S. Government Printing Office, Washington, DC.
- Naudet, J.P., Rigaud, P. and Huguenin, D. (1980) Stratospheric NO₂ at night from stellar spectra in the 440nm region, Geophys. Res. Letts; 7, 701-701. [NO₂]
- Oelhaf, H., Lenpolt, A. and Fischer, H. (1983) Discrepancies between balloon-borne IR atmospheric spectra and corresponding synthetic spectra calculated line-by-line around 825 cm⁻¹, Appl. Opt.; 21, 674-649. [NH₃]
- Philbrick, C.R. private communication based on Keneshea, T.J., Zimmerman, S.P. and Philbrick, C.R. (1979) A dynamic model of the mesosphere and lower thermosphere, Planet. Space. Sci.; 27, 285-401. [CO, O, O₂, O₃, H₂O, NO, OH]
- Rasmussen, R.A., Khalil, M.A.K., and Fox, R.J. (1983) Altitudinal & temporal variation of hydrocarbons and other gaseous tracers of arctic haze, Geophys. Res. Lett.; 10, 144-147. [C₂H₂, C₂H₆, C₂Cl₄]
- Remsberg, E.E., Russell III, J.M., Gille, J.C., Gordley, L.L., Bailey, P.L., Planet, W.G. and Harries, J.E. (1984a) The validation of Nimbus 7 LIMS measurements of ozone, J. Geophys. Res.; 89, 5161-5178. [O₃]
- Remsberg, E.E., Russell III, J.M., Gordley, L.L., Gille, J.C. and Bailey, P.L. (1984b) Implications of the stratospheric water vapor distribution as determined from the Nimbus 7 LIMS experiment, J. Atmos. Sci.; 41, 2934-2945. [H₂O]

- Rinsland, C.P., Goldman, A., Murcray, F.J., Murcray, D.G., Smith, M.A.H., Seals, R.K., Larsen, J.C. and Rinsland, P.L. (1982a) Stratospheric N_2O mixing ratio profile from high-resolution balloon-borne solar absorption spectra and laboratory spectra near 1880 cm^{-1} , Appl. Opt.; 21, 4351-4355. [N_2O]
- Rinsland, C.P., Smith, M.A.H., Rinsland, P.L., Goldman, A., Brault, J.W. and Stokes, G.M. (1982) Ground-based infrared spectroscopy measurements of atmospheric hydrogen cyanide, J. Geophys. Res.; 87, 11119-11125. [HCN]
- Rinsland, C.P., Boughner, R.E., Larsen, J.C., Stokes, G.M. and Brault, J.W. (1984) Diurnal variations of atmospheric NO: ground-based infrared spectroscopic measurements and their interpretation with time-dependent photochemical model calculations, J. Geophys. Res.; 89, 9613-9622. [NO]
- Rothman, L.S., Goldman, A., Gillis, J.R., Gamache, R.R., Pickett, H.M., Poynter, R.L., Husson, N. and Chedin, A. (1983b), AFGL trace gas compilation: 1982 version, Appl. Opt., 22, 1616-1627.
- Rothman, L.S., Gamache, R.R., Barbe, A., Goldman, A., Gillis, H.R., Brown, L.A., Toth, R.A., Flaud, J.-M. and Camy-Peyret, C. (1983a), AFGL atmospheric absorption line parameters compilation: 1982 edition, Appl. Opt., 22, 2247-2256.
- Rudolph, J., Ehhalt, D.H., Chedin, A. (1984) Vertical profiles of acetylene in the troposphere and stratosphere, J. Atmos. Chem., 2, 117-124. [C_2H_2]
- Russell III, J.M., Gille, J.C., Remsberg, E.E., Gordley, L.L., Bailey, P.L., Drayson, S.R., Fischer, H., Girard, A., Harries, J.E. and Evans, W.F.J. (1984a) Validation of nitrogen dioxide results measured by the LIMS experiment on Nimbus 7, J. Geophys. Res.; 89, 5099-5108. [NO_2]
- Russell III, J.M., Gille, J.C., Remsberg, E.E., Gordley, L.L., Bailey, P.L., Fischer, H., Girard, A., Drayson, S.R., Evans, W.F.J. and Harries, J.E. (1984b) Validation of water vapor results measured by the LIMS experiment on Nimbus 7, J. Geophys. Res.; 89, 5115-5120. [H_2O]
- Russell III, J.M., Solomon, S., Gordley, L.L., Remsberg, E.E. and Callis, L.B. (1984c) The variability of stratospheric and mesospheric NO_2 in the polar winter night observed by LIMS, J. Geophys. Res.; 89, 7267-7276. [NO_2]
- Seiler, W. and Fishman, J. (1981) The distribution of carbon monoxide and ozone in the free troposphere, J. Geophys. Res.; 86, 7255-7666. [CO , O_3]
- Smith, M.A.H. (1982) Compilation of atmospheric gas concentration profiles from 0-50km, NASA Tech Mem 83289, 70 pp. [all]
- Smith, M.A.H. and Rinsland, C.P. (1985), Spectroscopic measurements of atmospheric HCN at northern and southern latitudes, Geophys. Res. Lett., 12, 5-8. [HCN]
- Solomon, S., Crutzen, P.J. and Roble, R.G. (1982) Photochemical coupling between the thermosphere and the lower atmosphere: I. Odd nitrogen from 50-120 km, J. Geophys. Res.; 87, 7206-7221. [OH]
- Solomon, S., Garcia, R.R., Olivero, J.J., Bevilacqua, R.M., Schwartz, P.R., Clancy, R.T. and Mahleman, D.O. (1985) Photochemistry and transport of CO in the middle atmosphere, J. Atmos. Sci., 42, 1072-1083. [CO]
- Solomon, S., Mount, G.H. and Zawodny, J.M. (1984) Measurements of stratospheric NO_2 from the Solar Mesospheric Explorer (SME) satellite, 2, General morphology of observed NO_2 and derived N_2O_5 , J. Geophys. Res.; 89, 7317-7321. [NO_2]
- Stordal, F., Isaksen, I.S.A. and Hornveth, K. (1985) A diabatic circulation two-dimensional model with photochemistry: simulations of ozone and long-lived tracers with surface sources, J. Geophys. Res.; 90, 5757-5776. [CH_4 , N_2O , H_2O]

- Thomas, R.J., Barth, C.A., Rusch, D.W. and Sanders, R.W. (1984) Solar Mesospheric Explorer (SME) near-infrared spectrometer: Measurements of $1.27\mu\text{m}$ radiances and the inference of mesospheric ozone, J. Geophys. Res; 89, 9569-9580. [O₃]
- Thomas, R.J., Barth, C.A. and Solomon, S. (1984) Seasonal variations of O₃ in the upper mesosphere and gravity waves, Geophys. Res. Letts., 11, 673-676. [O₃]
- Turco, R.P., Hamill, P., Toon, O.B., Whitten, R.C. and Kiang, C.S. (1979) One-dimensional model describing aerosol formation and evolution in the stratosphere: I. Physical processes and mathematical processes and mathematical analogs, J. Atmos. Sci., 36, 699-717. [HSO₃, H₂SO₄, SO₂, OCS]
- Vukovich, F.M., Fishman, J. and Browell, E.V. (1985) The reservoir of ozone in the boundary layer of the Eastern United States and its potential impact on the global tropospheric ozone budget, J. Geophys. Res; 90, 5687-5698. [O₃]
- Wattenbach, R., Durwen, E.J., Roser, H.P. and Schultz, G.V. (1984) Observation of the CO (J=6+5) rotational transition in the earth's upper atmosphere, J. Geophys. Res; 89, 7285-7290. [CO]
- Weeks, L.H., Good, R.E., Randhawa, J.S. and Trinks, A. (1978) Ozone measurements in the stratosphere, mesosphere, and lower thermosphere during Aladdin 74, J. Geophys. Res; 83, 978-982. [O₃]
- Weinreb, M.P., Morgan, W.A., Chang, I-Lok, Johnson, L.D., Bridges, P.A. and Neuendorffer, A.C. (1984) High-altitude balloon test of satellite solar occultation instrument for monitoring stratospheric O₃, H₂O and HNO₃, J. Atmos. & Oceanic Tech., 1, 87-100. [H₂O, HNO₃, O₃]
- Weinstock, E.M., Phillips, M.J. and Anderson, J.G. (1981) In situ observations of ClO in the stratosphere: a review of recent results, J. Geophys. Res; 86, 7273-7278. [ClO]
- Weis, R.F. (1984) The temporal & spatial distribution of tropospheric nitrous oxide, J. Geophys. Res; 86, 7185-7196. [NO₂]
- W.M.O. (1982), The Stratosphere 1981: Theory and Measurements, Report No. 11, NASA, Greenbelt, MD.
- W.M.O. (1986), Atmospheric Ozone 1985: Assessment of our understanding of the processes controlling its present distribution and change, WMO Report No. 16, W.M.O., Geneva, Switzerland.